INCH-POUND

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SUPERSEDING
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DETAIL SPECIFICATION

CONNECTORS, ELECTRICAL, CIRCULAR, FOR HIGH-SPEED DATA BUS TRANSMISSION, COPPER CONDUCTOR, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements and test methods for high-speed data connectors using copper conductors for the transmission of signals over extended lengths. This specification contains connector definitions, contact arrangements and port assignments for single and multiple high-speed data transmission within a single connector shell.

1.2 Classification. Connectors are of the following series, classes, temperature ranges, and have the following contact styles, as specified (see 6.2).

1.2.1 Series. The connectors are series III, scoop-proof, triple start, self-locking, threaded coupling connectors in accordance with MIL-DTL-38999.

1.2.2 Classes. The connector classes are in accordance with MIL-DTL-38999, series III.

1.2.3 Temperature ranges. The connector temperature ranges are in accordance with MIL-DTL-38999, as defined for series III connector and finish. The upper temperature is the maximum internal hotspot temperature resulting from any combination of electrical load and ambient conditions.

Comments, suggestions, or questions on this document should be addressed to the Naval Air Warfare Center Aircraft Division Lakehurst, Code 4.1.2.2, Mail Stop 120-3, Route 547, Joint Base MDL, NJ 08733-5100 or emailed to michael.sikora@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at https://assist.dla.mil.

AMSC N/A FSC 5935

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1.2.4 Contact styles. Contact styles are as follows:

P – Pin - 500 cycle contact

S – Socket - 500 cycle contact

1.3 Part or Identifying number (PIN). The PIN to be used for connectors acquired to this specification is defined in the specification sheet (see 3.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARD

FED-STD-H28	-	Screw-Thread Standards for Federal Services

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-S-7742		Screw Threads, Standard, Optimum Selected Series: General Specification for
MS27488	-	Plug, End Seal, Electrical Connector
MIL-DTL-38999	-	Connectors, Electrical, Circular, Miniature, High Density, Quick Disconnect (Bayonet, Threaded, or Breech Coupling), Environment Resistant, with Crimp Removable Contacts or Hermetically Sealed with Fixed, Solderable Contacts, General Specification for

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MIL-DTL-38999/28	-	Connectors, Electrical, Circular, Nut, Hexagon, Connector Mounting, Series III and IV, Metric
MIL-A-46146	-	Adhesives-Sealants, Silicone, RTV, Non-corrosive (for use with Sensitive Metals and Equipment)
MIL-DTL-32546/1	-	Connectors, Electrical, Circular, Receptacle, High-speed Data, Threaded, Square Flange
MIL-DTL-32546/2	-	Connectors, Electrical, Circular, Plug, High-speed Data
MIL-DTL-32546/3	-	Connectors, Electrical, Circular, Receptacle, Threaded, Jam Nut Mounting, High-speed Data

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-202-110	-	Method 110, Sand and Dust
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-889	-	Dissimilar Materials
MIL-STD-1285	-	Marking of Electrical and Electronic Parts

(Copies of these documents are available online at http://quicksearch.dla.mil/.)

2.2 Other government documents. The following other Government document forms a part of this document to the extent specified herein. Unless otherwise specified, the issue of this document is that cited in the solicitation or contract.

DEFENSE STANDARDIZATION PROGRAM OFFICE

SD-6 - Provisions Governing Qualification

(Copies of this document are available online at http://quicksearch.dla.mil.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D570	-	Standard Test Method for Water Absorption of Plastics
ASTM E595	-	Standard Test Method for Total Mass Loss and Collected Volatile Condensable Materials from Outgassing in a Vacuum Environment

(Copies of these documents are available from http://www.astm.org.)

ELECTRONIC COMPONENTS INDUSTRY ASSOCIATION (ECIA)

ECIA EIA-359	-	EIA Standard Colors for Color Identification and Coding
ECIA EIA-364	-	Electrical Connector/Socket Test Procedures Including Environmental Classifications
ECIA EIA-364-03	-	Altitude Immersion Test Procedure for Electrical Connectors
ECIA EIA-364-10	-	Fluid Immersion Test Procedure for Electrical Connectors
ECIA EIA-364-14	-	Ozone Exposure Test Procedure for Electrical Connectors
ECIA EIA-364-20	-	Withstanding Voltage Test Procedure for Electrical Connectors, Sockets and Coaxial Contacts
ECIA EIA-364-21	-	Insulation Resistance Test Procedure for Electrical Connectors, Sockets, And Coaxial Contacts
ECIA EIA-364-23	-	Low Level Contact Resistance Test Procedure for Electrical Connectors and Sockets
ECIA EIA-364-24	-	Maintenance Aging Test Procedure for Electrical Connectors
ECIA EIA-364-26	-	Salt Spray Corrosion Test Procedure for Electrical Connectors, Contacts and Sockets
ECIA EIA-364-27	-	Mechanical Shock (Specified Pulse) Test Procedure for Electrical Connectors and Sockets

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ECIA EIA-364-28	-	Vibration Test Procedure for Electrical Connectors and Sockets
ECIA EIA-364-29	-	Contact Retention Test Procedure for Electrical Connectors
ECIA EIA-364-31	-	Humidity Test Procedure for Electrical Connectors and Sockets
ECIA EIA-364-32	-	Thermal Shock (Temperature Cycling) Test Procedure for Electrical Connectors and Sockets
ECIA EIA-364-35	-	Insert Retention Test Procedure for Electrical Connectors
ECIA EIA-364-42	-	Impact Test Procedure for Electrical Connectors
ECIA EIA-364-46	-	Microsecond Discontinuity Test Procedures for Electrical Connectors, Contacts and Sockets
ECIA EIA-364-54	-	Magnetic Permeability Test Procedure for Electrical Connectors, Contacts and Sockets
ECIA EIA-364-66	-	EMI Shielding Effectiveness Test Procedure for Electrical Connectors
ECIA EIA-364-75	-	Lightning Strike Test Procedure for Electrical Connectors
ECIA EIA-364-78	-	Cavity-to-Cavity Leakage Bonding Integrity Test Procedure for Electrical Connectors
ECIA EIA-364-79	-	Insert Bond Strength Test Procedure for Electrical Connectors
ECIA EIA-364-83	-	Shell-To-Shell and Shell-To-Bulkhead Resistance Test Procedure for Electrical Connectors
ECIA EIA-364-90	-	Crosstalk Ratio Test Procedure for Electrical Connectors, Sockets, Cable Assemblies or Interconnect Systems
ECIA EIA-364-101	-	Attenuation Test Procedure for Electrical Connectors, Sockets, Cable Assemblies or Interconnection Systems

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ECIA EIA-364-103	-	Propagation Delay Test Procedure for Electrical Connectors, Sockets, Cable Assemblies or Interconnect Systems
ECIA EIA-364-105	-	Altitude - Low Temperature Test Procedure for Electrical Connectors and Sockets
ECIA EIA-364-108	-	Impedance, Reflection Coefficient, Return Loss, and VSWR measured in the Time and Frequency Domain Test Procedure for Electrical Connectors, Cable Assemblies or Interconnection Systems
ECIA EIA-364-114	-	Coupling and Uncoupling Force Test Procedure for Electrical Connectors, Sockets, and Applicable Accessories

(Copies of these documents are available online at http://www.ecianow.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 68-1	-	General Purpose Screw Threads, - Basic Profile, Part 1: Metric Screw Threads
ISO 261	-	General Purpose Metric Screw Threads, General Plan
ISO 262	-	General Purpose Metric Screw Threads - Selected Sizes for Screws, Bolts and Nuts
ISO 965-1	-	General Purpose Metric Screw Threads - Tolerances - Part 1: Principles and Basic Data
ISO 965-2	-	General Purpose Metric Screw Threads - Tolerances - Part 2: Limits of Sizes for General Purpose External and Internal Screw Threads - Medium Quality
ISO 965-3	-	General Purpose Metric Screw Threads - Tolerances - Part 3: Deviations for Constructional Screw Threads

(Copies of these documents are available online at http://www.iso.org or http://webstore.ansi.org/.)

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SAE INTERNATIONAL

SAE AIR1351	-	Manufacturers' Identification of Aerospace Electrical and Electronic Wiring Devices and Accessories
SAE AS3582	-	Packing, Preformed – O-Ring Seal AMS3304
SAE AS6070	-	Aerospace Cable, High Speed Data, Copper
SAE AS22520	-	Crimping Tools, Wire Termination, General Specification for
SAE AS39029	-	Contacts, Electrical Connector, General Specification for
SAE AS81969/8	-	Installing and Removal Tools, Connector Electrical Contact, Type I and II, Class 2, Composition A
SAE AS81969/14	-	Installing and Removal Tools, Connector Electrical Contact, Type III, Class 2, Composition B
SAE AS81969/16	-	Installing and Removal Tools, Connector Electrical Contact, Type III, Class 1, Composition B
SAE AS85049	-	Connector Accessories, Electrical, General Specification for
SAE AS85049/80	-	Connector Accessories, Electrical, Dummy Contact, Sizes 16, 12 and 8, Category 7 (For MIL-DTL-38999 Connectors)
SAE AS85049/81	-	Connector Accessories, Electrical, Seal Plug, Size 10, Category 7 (For MIL-DTL-38999 Connectors)
SAE AS85049/103	-	Connectors, Accessories, Composite, RFI/EMI, Electrical, Strain Relief, Straight, Self-Locking, Category 3C (for MIL-DTL-38999 Series III and IV Connectors)
SAE AS85049/128	-	Connector Accessories, Electrical, Backshell, Shield Band, Category 7 (For AS85049/82-/90, /93, /109-/117 Accessories.)

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(Copies of these documents are available online from http://www.sae.org.)

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-568-C.2	-	Balanced Twisted-Pair Telecommunications Cabling and Components Standard

(Copies of these documents are available online at http://www.tiaonline.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern.

3.2 Cable. Connectors shall be compatible with SAE AS6070 cables. Specific SAE AS6070 cable specification sheets shall be defined in the applicable MIL-DTL-32546 specification sheets.

3.3 Qualification. Connectors furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.3 and 6.3).

3.3.1 Physical characteristics. Connectors shall meet the requirements of series III as defined in MIL-DTL-38999.

3.3.2 Size. Unless otherwise specified, connector shell sizes shall be in accordance with MIL-DTL-38999.

3.3.3 Crimp removable contact. Crimp removable contacts shall be used. Unless otherwise specified, crimp contacts shall be qualified products in accordance with SAE AS39029. All connector classes shall make provision for individual contact removal through the use of a contact extraction tool. Unless otherwise specified, the contacts shall be rear release. The quantity of crimp contacts to be supplied with each connector unit package shall consist of a full complement of contacts plus 1 spare contact for each size used in the arrangement utilizing 26 contacts or less. For arrangements utilizing more than 26 contacts, 2 spare contacts of each size used in the arrangement shall be supplied.

3.3.4 Electrical performance. All connector classes shall be of a design and construction such that individual pairs of contacts shall meet the following electrical characteristics for the frequency range and electrical performance requirements defined in this specification and the individual detail specification sheets.

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3.3.4.1 Nominal impedance. All connector classes shall be of a design and construction such that individual pairs of contacts shall maintain nominal impedance.

3.3.4.2 Insertion loss. The signal loss resulting from the insertion of the connector between the signal generation and detection device shall be less than the values defined in TIA-568-C.2 insertion loss under connecting hardware transmission performance for the frequency range specified in the detail specification sheet. (See 4.3.5.1).

3.3.4.3 Return loss. The ratio of the transmitted signal power relative to the reflected signal level shall meet or exceed the values defined in TIA-568-C.2 return loss under connecting hardware transmission performance for the frequency range specified in the detail specification sheet. (See 4.3.5.2).

3.3.4.4 Near-end crosstalk. Noise induced by the active data pair to an adjacent pair measured at the near end relative to active signal level shall meet or exceed the values defined in TIA-568-C.2 near-field crosstalk, under connecting hardware transmission performance for the frequency range specified in the detail specification sheet. (See 4.3.5.3).

3.3.4.5 Far-end crosstalk. Noise induced by the active data pair to an adjacent pair measured at the far end relative to active signal level shall meet or exceed the values defined in TIA-568-C.2 far-end crosstalk, under connecting hardware transmission performance for the frequency range specified in the detail specification sheet. (See 4.3.5.4).

3.3.4.6 Propagation delay. The time delay for a signal to travel from one end of a mated connector pair to the opposite end for each data pair within the connector shall not exceed 2.5 nanoseconds (ns) from 1 megahertz (MHz) to the maximum frequency range specified in the detail specification sheet. (See 4.3.5.5).

3.3.4.7 Propagation delay skew. The difference in time between the travel time of the fastest and slowest data pair within a connector data channel shall not exceed 1.25 ns from 1 MHz to the maximum frequency range specified in the detail specification sheet. (See 4.3.5.6).

3.4 Materials.

3.4.1 Critical interface materials. Materials shall be as specified herein. If materials other than those specified herein are used, the contractor shall certify to the qualifying activity (see 6.3) that the substitute material enables the connectors to meet the requirements of this specification. The supplier shall request authorization from the qualifying activity to use a substitute material and the qualifying activity reserves the right to require full or partial qualification testing. Acceptance or approval of any constituent material shall not be construed as a guaranty of acceptance of the product. When a definite material is not specified, a material shall be used which will enable the connector to meet the requirements of this specification.

3.4.1.1 Metals. Metals shall be of a corrosion-resistant type or shall be plated or treated to resist corrosion.

3.4.1.2 Dissimilar metals and compatible couples. When dissimilar metals are used in intimate contact with each other, protection against galvanic corrosion shall be provided. The use of dissimilar metals in contact, which tend toward active galvanic corrosion (particularly brass, copper, or steel used in contact with aluminum or an aluminum alloy), is not acceptable. However, metal plating of dissimilar base metals to provide similar or suitable abutting surfaces is permitted. The use of dissimilar metals separated by a suitable insulating material is also permitted. Dissimilar metals and compatible couples are specified in MIL-STD-889.

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3.4.2 Components. Materials for specific components of the connector shall be in accordance with the components paragraph of MIL-DTL-38999.

3.4.3 Magnetic permeability. The relative permeability of the terminated, assembled and fully mated connector assembly shall be less than 2.0 microns (μ) when measured in accordance with 4.6.32.

3.4.4 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4.5 Plating.

3.4.5.1 Contacts. Contacts shall be in accordance with SAE AS39029 (see 3.3.3) and the contact plating shall be in accordance with the applicable SAE AS39029 contact specified in the detail specification sheet.

3.4.5.2 Shells and accessory hardware. Unless otherwise specified, the finish on shells and accessory hardware shall be in accordance with the shells and accessory hardware paragraph of MIL-DTL-38999 for series III connectors.

3.4.5.3 Shell spring fingers. Shell spring fingers shall be suitably protected to prevent corrosion.

3.5 Configuration interface and features. Connectors and accessories shall be configured to withstand normal handling incident to installation and maintenance in service. Connector intermateability control dimensions shall be designed to meet the requirements of MIL-DTL-38999, series III.

3.5.1 Contact arrangement. Contact arrangement shall be as specified in the detail specification sheet. The engaging end of the pin contact in assembled connectors shall be located within .024 inch (0.61 mm) diameter of true position and the engaging end of socket contacts in assembled connectors shall be located within .015 inch (0.38 mm) diameter of true position. Test voltages shall be in accordance with Table I.

TABLE I. Test voltages, AC rms*, 60 Hertz (Hz).

Altitude	Mated	Unmated
Sea level	1300	1300
50,000 feet	800	550
70,000 feet	800	350
100,000 feet	800	200

*rms – root mean square

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3.5.2 Installing and removal tools. SAE AS81969/8, SAE AS81969/14 and SAE AS81969/16 tools shall be used to properly assemble and disassemble the pin and socket contacts into their connector inserts. One SAE AS81969/14 or one SAE AS81969/16 tool for each contact size as applicable shall be enclosed in the unit package. For indirect shipments, connectors may be specified without installation and removal tools.

3.5.3 Backshell provision. Connectors shall include provisions for termination of cable braid. Connectors shall be compatible with SAE AS85049 backshells to permit shield termination using M85049/128-5, 6, 7, or 8 bands per SAE AS85049/128.

3.5.4 Dust caps/containers. Dust caps shall be placed on the mating end of each connector.

3.5.5 Insert interface. The entire insert and wire sealing or wire supporting member of the environment resisting assemblies shall be essentially one integral part, configured to provide suitable sealing and support (except box mount connectors) around the wires and shall be non-removable. The rigid dielectric shall be one molded piece or no more than two pieces bonded so as to form one integral piece. The configuration shall be such as to permit the removal and replacement of individual contacts into their connector inserts with either SAE AS81969/8 or SAE AS81969/14 installation/removal tools. The contact locking device shall be contained in the rigid dielectric insert and shall so retain the contacts as to meet the contact retention requirements of this specification. Inserts shall be secured to prevent rotation. All pin contact inserts shall have a resilient interface seal bonded to the front face in accordance with the applicable standards. Socket entry holes and pin "donut" rings shall conform to the requirements of MIL-DTL-38999 (see the insert interface paragraph of MIL-DTL-38999).

3.5.6 Sealing.

3.5.6.1 Wire sealing. Not applicable to box mount connectors. Environment resisting assemblies shall be designed to meet the environmental requirements of this specification using wire of outer diameter within the applicable range. (See Table II).

TABLE II. Wire and cable sizes and diameters.

Wire size (AWG*)	finished wire outside dimensions			
	minimum		maximum	
	inch	mm	inch	mm
28, 26, 24, 22	0.034	0.86	0.060	1.52

*AWG – American Wire Gauge

NOTE: Connectors shall meet the requirements specified when:

a. A full complement of wire of the applicable minimum or maximum insulation diameter is installed.

b. Any combination of wire diameters within the extremes of (a) above are used.

3.5.6.2 Grommet sealing plugs. Except for box mount connectors and as otherwise specified, the grommets of environment resisting connectors shall be designed to accept sealing plugs/dummy contacts in accordance with MS27488, SAE AS85049/80 or SAE AS85049/81 as applicable, in lieu of wire where unwired contacts are employed. The quantity of sealing plugs to be enclosed in the unit pack shall be ten percent of the number of contacts, but not less than four. For indirect shipments, connectors may be specified without grommet sealing plugs.

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3.5.6.3 Mating seal. Plugs and receptacles with pin inserts shall have a resilient face with individual pin barriers (see the mating seal paragraph of MIL-DTL-38999). The pin barrier projections shall seal in their respective lead-in chambers of the hard face socket insert. The resilient interfacial seal shall provide individual contact seals in the mated condition to ensure circuit isolation between each contact and contact to shell. The receptacles shall be provided with a peripheral seal.

3.5.7 Shell. Shells, including mounting flanges, shall be of one-piece construction and shall be designed to retain their inserts in one position, both axially and with respect to rotation, by mechanical means. Adhesive may be used as a supplementary retention means for environment resisting connectors. Each plug and receptacle connector shall have at least one blue color band in accordance with ECIA EIA-359, indicating a rear release contact retention system. The blue color band shall be located so that it is readily visible to any person servicing a mounted connector.

3.5.7.1 Spring fingers. All plugs shall have spring fingers. Spring fingers shall be designed to make electrical contact with the mating shell without interfering with proper engagement. The springs shall be retained about the shell periphery. Minimum engagement of spring fingers shall be 0.040 inch (1.02 mm) prior to contact engagement.

3.5.7.2 Jam nut mounting receptacles. Jam nut mounting receptacles shall be provided with a mounting nut in accordance with MIL-DTL-38999/28 with provisions for locking, and an O-ring in accordance with SAE AS3582.

3.5.8 Screw threads. Screw threads shall conform to FED-STD-H28, MIL-S-7742, ISO 68-1, ISO 261, ISO 262, ISO 965-1, ISO 965-2, and ISO 965-3 as applicable. Threads shall be checked using ring or plug gauges. Slight out-of-roundness beyond the specified tolerances is acceptable if threads can accept the gauges without forcing.

3.5.9 Coupling. Connectors shall be coupled to counterpart connectors by means of a triple start thread mechanism. The mechanism shall include a means of maintaining the mated connector in full engagement. The coupling ring shall be knurled or fluted to facilitate coupling and shall be captivated. The coupling rings of all connectors shall have a blue color band in accordance with ECIA EIA-359, indicating a rear release contact retention system. The end surfaces of all bayonet pins shall be of contrasting blue color to the shell finish.

3.5.9.1 Ease of coupling. Counterpart connectors of any arrangement shall be capable of being fully coupled and uncoupled in a normal and accessible location without the use of tools.

3.5.9.2 Locking. Complete coupling shall be accomplished by approximately 360° clockwise rotation of the coupling nut and shall provide shell-to-shell bottoming when tested in accordance with 4.6.14.1.2. Mated connectors, when vibrated, shall not display any resonance at frequencies less than 300 Hz. An anti-decoupling device shall be provided to maintain complete coupling. A solid red band, uniform in color, shall be located on the receptacle so as to be visible when unmated and fully covered when completely mated.

3.5.9.3 Polarization of connector shells. Polarization of connector shells shall be accomplished by means of integral keys and suitable matching keyways on the counterpart connector. Polarization shall be accomplished before initial engagement of the coupling ring. During axial engagement, pins shall not touch sockets or the insert face until polarization has been achieved.

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3.5.9.4 Lubrication. Coupling threads of connectors may be coated with a suitable lubricant. Lubricants shall be non-flaking and galvanically compatible, and shall not migrate into threaded areas. Rear accessory threads of all connectors and receptacle coupling areas shall not be lubricated.

3.5.9.5 Pin-to-pin mating prevention. Connectors shall be designed such that pin-to-pin contact (electrical or physical) is not possible in the event that a plug with pin contacts is inadvertently mated with a receptacle with pin contacts.

3.5.10 Cavity fills. If the rear grommet design does not allow for intimate contact between it and the complete inner perimeter of the shell, any resulting cavity between the insert and the shell shall be filled with RTV silicone conforming to MIL-A-46146, or equivalent. The silicone shall adhere to both the shell and the grommet.

3.6 Interchangeability. All connectors having the same Part or Identifying Number (PIN) shall be completely interchangeable with each other with respect to installation and performance.

3.7 Maintenance aging. When tested as specified in 4.6.1, the contact installing and removal forces shall not exceed the requirements in Table III.

TABLE III. Installing and removal forces.

Contact size	Installing and removal forces (maximum)	
	pounds	Newtons
22D, 23-22	10	44

3.8 Temperature cycling. When tested as specified in 4.6.2 there shall be no blistering, peeling, flaking or separation plating or other damage detrimental to the operation of the connectors.

3.9 Plating adhesion (applicable to classes J and M only). When tested as specified in 4.6.34, there shall be no blistering, peeling, flaking or separation of plating or other damage detrimental to the operation of the connector.

3.10 Coupling torque. When tested as specified in 4.6.3, the coupling torque for mating and unmating of the counterpart connectors shall meet the requirements in Table IV.

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TABLE IV. Coupling torque.

Shell size	Maximum engagement and disengagement		Minimum disengagement	
	Pound inch	Newton meters	Pound inch	Newton meters
9	8	0.9	2	0.2
11	12	1.4	2	0.2
13	16	1.8	2	0.2
15	20	2.3	3	0.3
17	24	2.7	3	0.3
19	28	3.2	3	0.3
21	32	3.6	5	0.6
23	36	4.1	5	0.6
25	40	4.6	5	0.6

3.11 Durability. When tested as specified in 4.6.4, the connectors shall show no defects detrimental to the operation of the connectors and shall meet the subsequent test requirements.

3.12 Altitude immersion. When tested as specified in 4.6.5, the mated connector pair shall have a minimum insulation resistance as specified in 3.13.1 and dielectric withstanding voltage as specified in 3.14.

3.13 Insulation resistance (see 4.6.6).

3.13.1 Insulation resistance at ambient temperature. When tested as specified in 4.6.6.1, the insulation resistance between any pair of contacts and between any contact and the shell shall be greater than 5,000 megohms minimum. Insulation resistance after altitude immersion shall be 1,000 megohms minimum. Insulation resistance after humidity shall be 100 megohms minimum.

3.13.2 Insulation resistance at elevated temperature. When tested as specified in 4.6.6.2, the insulation resistance between any pair of contacts and between any contact and the shell shall be greater than 1,000 megohms minimum at 175 ^{0}C for environment resisting class connectors.

3.14 Dielectric withstanding voltage. When tested as specified in 4.6.7, the maximum leakage current shall be 2 milliamps, and there shall be no evidence of electric breakdown or flashover.

3.15 Insert retention. When tested as specified in 4.6.8, unmated connectors shall retain their proper location in the shell and there shall be no evidence of cracking, breaking,

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separation from the shell, or loosening of parts. The insert retention shall be a minimum of 25 pounds force (lbf).

3.16 Salt spray (corrosion). When tested as specified in 4.6.9, unmated connectors shall show no lifting of plated coating or exposure of basis material under three power (3X) magnification which adversely affects performance when evaluated in accordance with ECIA EIA-364-26.

3.17 Low level contact resistance. When tested as specified in 4.6.10, (see SAE AS39029 type A contacts, silver plated wire, after conditioning), the maximum low level contact resistance shall be 30 milliohms.

3.18 Electrical engagement. When tested as specified in 4.6.11, wired, mated connectors shall provide a minimum of 0.050 inch (1.27mm) electrical engagement.

3.19 Contact retention. When tested as specified in 4.6.12, the contact retention force shall be a minimum of 6 lbf with a maximum displacement of 0.012 inch.

3.20 Attitude-low temperature (100,000 feet, -65 °F). When tested as specified in 4.6.13, the mated connector pair shall have a minimum insulation resistance as specified in 3.13.1.

3.21 Vibration. When tested as specified in 4.6.14, there shall be no electrical discontinuity and there shall be no disengagement of the mated connectors, backing off of the coupling mechanism, evidence of cracking, braking or loosening of parts.

3.22 Shock. When tested a specified in 4.6.15, there shall be no electrical discontinuity and there shall be no disengagement of the mated connectors, evidence of cracking, breaking, or loosening of parts.

3.23 Shell-to-shell conductivity. When tested as specified in 4.6.16, the shell-to-shell conductivity shall not exceed 5 millivolts (mV) voltage drop initially and 10 mV voltage drop final. The voltage drop across the mated connectors shall be measured using 1 ampere across the connectors.

3.24 Backshell shield braid-to-shell conductivity. When tested as specified in 4.6.17, the braid-to-shell conductivity shall not exceed 1 mV voltage drop. The voltage drop across the mated connectors shall be measured using 1 ampere across the connector and braid.

3.25 Humidity. When tested as specified in 4.6.18, wired, mated connectors shall show no deterioration that will adversely affect performance of the connector. During the final cycle, insulation resistance shall be 100 megohms or greater.

3.26 Shell spring finger forces. When tested as specified in 4.6.19, the forces necessary to engage and separate electromagnetic interference (EMI) plugs with receptacle shells shall be within the values specified in Table V.

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TABLE V. Shell spring finger forces.

Shell size	Axial force			
	Maximum		Minimum	
	Pounds	Newtons	Pounds	Newtons
9	25	111	0.5	2
11	25	111	0.5	2
13	30	133	0.5	2
15	30	133	0.5	2
17	35	156	0.5	2
19	35	156	0.5	2
21	35	156	0.5	2
23	35	156	0.5	2
25	35	156	0.5	2

3.27 EMI shielding. When tested as specified in 4.6.20, the EMI shielding capabilities of mated shells with spring fingers shall not be less than that specified in MIL-DTL-38999 for series III connectors (see the EMI shielding paragraph in MIL-DTL-38999) at the specified frequencies.

3.28 Ozone exposure. When tested as specified in 4.6.21, connectors shall show no evidence of cracking of dielectric material or other damage due to ozone exposure that will adversely affect performance.

3.29 Fluid immersion. When tested as specified in 4.6.22, connectors shall meet the requirements for coupling torque and dielectric withstanding voltage.

3.29.1 Retention system fluid immersion. When tested as specified in 4.6.22.1, insert assemblies shall meet requirements of contact retention (see 3.19). Effects of the fluids on resilient sealing members shall not be a consideration during this test.

3.30 Pin contact stability. When tested as specified in 4.6.23, the total displacement of a reference point on the contact tip end shall not exceed the amount shown in Table VI.

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TABLE VI. Pin contact stability.

Contact size	Total displacement		Force	
	inch	mm	pounds	Newtons
23-22	0.025	0.64	0.25	1.1
22D	0.030	0.76	0.28	1.2

3.31 Contact walkout. When tested as specified in 4.6.24, contacts shall not become dislodged from their normal position.

3.32 Installing and removal tool abuse (applicable only to initial qualification). When tested as specified in 4.6.25, there shall be no evidence of damage to the contacts, the connector insert, or the contact retaining mechanism.

3.33 High temperature exposure. When tested as specified in 4.6.26, for 1,000 hours at 175 °C, connectors shall perform satisfactorily and pass succeeding tests in the qualification test sequence.

3.34 Electrolytic erosion. When tested as specified in 4.6.27, pin contacts shall show no exposure of base metal due to electrolytic erosion. Corrosion deposits shall not be considered as defects.

3.35 Ice resistance. When tested as specified in 4.6.28, connectors shall pass succeeding tests in Table VII. When tested in accordance with 4.6.3, uncoupling and recoupling torque shall not exceed the values specified in MIL-DTL-38999 (see coupling torque paragraph in MIL-DTL-38999) by 25 percent.

3.36 Dust (fine sand). When tested as specified in 4.6.29, connectors shall pass succeeding tests specified in Table VII. When tested as specified in 4.6.3, uncoupling and recoupling torque shall not exceed the values specified in MIL-DTL-38999 (see the coupling torque paragraph in MIL-DTL-38999) by more than 25 percent.

3.37 Cavity-to-cavity leakage bonding integrity. When tested as specified in 4.6.30, there shall be no air leakage between contact cavities.

3.38 Resistance to indirect lightning strike. When tested as specified in 4.6.31, wired and mated connectors shall meet braid shield braid-to-shell conductivity as specified in 3.24, coupling and uncoupling torque as specified in Table IV, insulation resistance at ambient temperature as specified in 3.13.1 and dielectric withstanding voltage at sea level as specified in 3.14. Connectors shall show no damage or degradation in the finish or base material that would affect subsequent use. Damage or hardening of elastomeric sealing members affecting sealing shall be considered a failure.

3.39 Marking. Connectors shall be permanently marked with PIN, date code, and the manufacturer's identification marking. The lot number (also called "lot/date code") shall be marked in accordance with MIL-STD-1285. The manufacturer's identification marking shall include either the manufacturer name or the Commercial and Government Entity (CAGE) Code and may also be identified by a unique identifier, such as a trademark, a symbol, a mint mark, or identification letters. The manufacturer's name, if used, may be abbreviated. If the manufacturer has more than one manufacturing facility, then the CAGE code of the product line shall be included on the part. If products are assembled by a qualified assembly plant, then the identification marking shall also include the assembly plant's identification marking. The

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trademark shall be as specified in SAE AIR1351 or shall be that which has been provided to the qualifying activity. Stamping shall be in accordance with MIL-STD-1285 where space permits. Metal stamping shall be accomplished before plating. A single digit shell size and/or insert arrangement shall be marked with a single digit (example: shell size 9 shall be marked "9" not "09").

3.40 Contact location identification. Contact locations shall be identified as specified on the applicable detail specification sheet. All positions shall be identified on the front and rear faces of the insert except where space limitations make this impracticable. Location of contact identifying characters shall be in close proximity to the holes but need not be placed exactly where specified on the specification sheet. The preferred color of the contact identifying character is white. When the background is a color against which white is difficult to distinguish, a color shall be chosen for which the identifying character can be easily distinguished.

3.41 Workmanship. The connector shall be fabricated in a manner such that the criteria for appearance, fit and adherence to specified tolerances are observed. Particular attention shall be given to neatness and thoroughness of marking parts, plating, welding, soldering, riveting, staking, and bonding. Visual examination under three power (3X) magnification shall include examination of each retention cavity for molding discrepancies, retention features, and excessive adhesive. The connectors shall be free from crazing, cracks, voids, pimples, chips, blisters, pinholes, sharp cutting edges, burrs, and other defects that will adversely affect life, serviceability, or appearance.

3.42 Assembly instructions. If specified in the contract (see 6.2), complete assembly instructions are furnished by the vendor with each connector procured under this specification. It is not the intention of this specification to require assembly instructions with uncabled connectors (i.e., solder pot, solder tab or posts, etc.). Assembly instructions include:

a. Cable preparation – stripping dimensions and tolerances.

b. List and description of crimping or special tools if required.

c. Pictorial presentation of sub-assemblies and loose piece parts.

d. Sufficient pertinent dimensions for verification of correct parts; as a minimum, the cable entry openings for conductor, dielectric, braid, and jacket shall be specified.

e. Recommended cable clamp tightening torque (if applicable).

f. Military PIN and manufacturer's PIN.

3.43 Gauge location. Using test gauges conforming to the requirements of MIL-DTL-38999 (see gauge location paragraph in MIL-DTL-38999), the axial location of contacts shall be measured as specified in 4.6.35. Gauge location measurements shall fall within the range specified in MIL-DTL-38999 (see gauge location paragraph in MIL-DTL-38999).

3.44 Gauge retention. When tested as specified in 4.6.36, the test gauges conforming to the dimensions specified in MIL-DTL-38999 (see gauge location paragraph in MIL-DTL-38999) shall be retained in the contact cavities of crimp contact connectors and the axial displacement of the test gauges while under load shall not exceed 0.012 inch (0.30 mm).

3.45 Fungus resistance. Materials used in the construction of these connectors shall be fungus inert (see 4.6.37).

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3.46 Insert bond strength. When tested as specified in 4.6.38, there shall be no visible separation or cracks at the bond joint while an axial load is applied.

3.47 Thermal vacuum outgassing (applicable only to classes G and H for initial qualification). All nonmetallic materials used in the finished connector shall not release greater than 1.0 percent total mass loss (TML) and 0.1 percent collected volatile condensable material (CVCM), when tested as specified in 4.6.39.

3.48 Impact (applicable only to classes J and M). When connector plugs with SAE AS85049 straight strain relief clamps are tested as specified in 4.6.40, there shall be no breaking or cracking of inserts or shells. Also, there shall be no bending of contacts nor any damage which would prevent the connector from mating or unmating.

3.49 Hydrolytic stability. Applicable to classes J and M only. For initial qualification only. When tested as specified in 4.6.41, the connector shall be without defects detrimental to mechanical performance. There shall be no increase in the connector weight greater than 0.75 percent. When subjected to an over torque at 150 percent of those values specified in Table IV, there shall be no evidence of cracking, breaking, or loosening of parts.

3.50 Signal integrity. When tested in accordance with 4.3.5, the connector shall meet 3.3.4.2, 3.3.4.3, 3.3.4.4, 3.3.4.5, 3.3.4.6, and 3.3.4.7.

4. VERIFICATION

4.1 Classification of inspection. The examinations and tests of connectors are classified as follows:

a. Qualification inspection (see 4.3).

b Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the “GENERAL REQUIREMENTS” of ECIA EIA-364.

4.2.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspection shall be established and maintained by the contractor.

4.3 Qualification inspection. Qualification inspection shall consist of the examinations and tests in Table VII. Unless otherwise specified in Table VII, sequential testing is required for each group. The qualifying activity shall perform or witness the particular tests as specified in Table VII and the supplier shall perform the remaining tests.

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TABLE VII. Qualification and qualification retention tests.

Inspection Group 1 (all classes and finishes) 1/	Requirement paragraph	Test paragraph
Visual and mechanical examination	3.1, 3.4, 3.5, 3.6, 3.39 and 3.41	4.5.1
Magnetic permeability	3.4.3	4.6.32
Maintenance aging	3.7	4.6.1
Temperature cycling	3.8	4.6.2
Coupling torque	3.10	4.6.3
Durability	3.11	4.6.4
Altitude immersion 2/	3.12	4.6.5
Insulation resistance at ambient temperature	3.13.1	4.6.6.1
Dielectric withstanding voltage at sea level	3.14	4.6.7.1

See footnotes at end of table.

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TABLE VII. Qualification and qualification retention tests – Continued.

Inspection Group 1 – continued (all classes and finishes) 1/	Requirement paragraph	Test paragraph
Insert retention	3.15	4.6.8
Shell-to-shell conductivity (not applicable to class C and finish C)	3.23	4.6.16
Salt spray (corrosion) (standard test applicable to classes F, G, N only)	3.16	4.6.9
Contact Resistance	3.17	4.6.10
Coupling torque	3.10	4.6.3
Shell-to-shell conductivity (not applicable to class C and finish C)	3.23	4.6.16
Electrical engagement	3.18	4.6.11
Post test examination	3.39, 3.40 and 3.41	4.6.33
Group 2 (all classes and finishes) 3/		
Visual and mechanical examination	3.1, 3.4, 3.5, 3.39 and 3.41	4.5.1
Gauge location	3.43	4.6.35
Gauge retention	3.44	4.6.36
Maintenance aging 3/	3.7	4.6.1
Contact retention	3.19	4.6.12
Altitude-low temperature 3/	3.20	4.6.13
Insulation resistance at ambient temperature	3.13.1	4.6.6.1
Dielectric withstanding voltage at sea level	3.14	4.6.7.1

See footnotes at end of table.

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TABLE VII. Qualification and qualification retention tests – Continued.

Inspection	Requirement paragraph	Test paragraph
Group 2 – continued (all classes and finishes) 3/		
Temperature cycling	3.8	4.6.2
Coupling torque	3.10	4.6.3
Insulation resistance at elevated temperature	3.13.2	4.6.6.2
Dielectric withstanding voltage at sea level	3.14	4.6.7.1
Dielectric withstanding voltage at altitude 2/	3.14	4.6.7.2
Durability	3.11	4.6.4
Vibration 4/	3.21	4.6.14
Low level contact resistance	3.17	4.6.10
Shock 3/	3.22	4.6.15
Low level contact resistance	3.17	4.6.10
Shell-to-shell conductivity (except finish C and class C)	3.23	4.6.16
High temperature exposure 2/	3.33	4.6.26
Humidity 3/	3.25	4.6.18
Insulation resistance at ambient temperature	3.13.1	4.6.6.1
Dielectric withstanding voltage at sea level	3.14	4.6.7.1
Contact retention	3.19	4.6.12
Post test examination	3.39, 3.40 and 3.41	4.6.33

See footnotes at end of table.

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TABLE VII. Qualification and qualification retention tests – Continued.

Inspection	Requirement paragraph	Test paragraph
Group 3 (shells with spring fingers) (all classes and finishes, except as noted below)		
Visual and mechanical examination	3.1, 3.4, 3.5, 3.6, 3.39 and 3.41	4.5.1
Durability (except classes J and M)	3.11	4.6.4
Shell spring finger forces	3.26	4.6.19
Shell-to-shell conductivity (except finish C and class C)	3.23	4.6.16
Salt spray (dynamic test) (classes J and M only)	3.16	4.6.9.2
Coupling torque (classes J and M only)	3.10	4.6.3
Shell-to-shell conductivity (classes J and M only)	3.23	4.6.16
EMI shielding (except finish C and class C)	3.27	4.6.20
Post test examination	3.39, 3.40 and 3.41	4.6.33
Group 4 (dielectric) (all classes and finishes)		
Visual and mechanical examination	3.1, 3.4, 3.5, 3.6, 3.39 and 3.41	4.5.1
Ozone exposure	3.28	4.6.21
Insulation resistance at ambient temperature	3.13.1	4.6.6.1
Dielectric withstanding voltage at sea level	3.14	4.6.7.1
Fluid immersion 2/	3.29	4.6.22
Dielectric withstanding voltage at sea level	3.14	4.6.7.1
Coupling torque	3.10	4.6.3

See footnotes at end of table.

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TABLE VII. Qualification and qualification retention tests – Continued.

Inspection	Requirement paragraph	Test paragraph
Group 4 (dielectric) - continued (all classes and finishes)		
Post test examination	3.39, 3.40 and 3.41	4.6.33
Group 5 (retention system) (all classes and finishes)		
Visual and mechanical examination	3.1, 3.4, 3.5, 3.6, 3.39 and 3.41	4.5.1
Retention system fluid immersion	3.29.1	4.6.22.1
Contact retention	3.19	4.6.12
Post test examination	3.39, 3.40 and 3.41	4.6.33
Group 6 (retention system) (all classes and finishes)		
Visual and mechanical examination	3.1, 3.4, 3.5, 3.6, 3.39 and 3.41	4.5.1
Pin contact stability	3.30	4.6.23
Contact walkout	3.31	4.6.24
Installing/removal tool abuse	3.32	4.6.25
High temperature exposure with contact loading	3.33	4.6.26.1
Insert retention	3.15	4.6.8
Post test examination	3.39, 3.40 and 3.41	4.6.33
Group 7 (classes C, F, G, H, J, K , L, N, M, R, S, T, U, W, X, Y and Z) 5/		
Visual and mechanical examination	3.1, 3.4, 3.5, 3.6, 3.39 and 3.41	4.5.1
Shock (high impact)	3.22	4.6.15.2
Dielectric withstanding voltage (at sea level)	3.14	4.6.7.1
Electrolytic erosion	3.34	4.6.27
Shell-to-shell conductivity (not applicable to class C and finish C)	3.23	4.6.16

See footnotes at end of table.

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TABLE VII. Qualification and qualification retention tests – Continued.

Inspection	Requirement paragraph	Test paragraph
Group 7 - continued (classes C, F, G, H, J, K , L, N, M, R, S, T, U, W, X, Y and Z) 5/		
Temperature cycling	3.8	4.6.2
Salt spray (corrosion) (dynamic test applicable to classes C,H,J,K,L,M,T,W,X,Y and Z)	3.16	4.6.9
Coupling torque	3.10	4.6.3
Shell-to-shell conductivity (not applicable to class C and finish C)	3.23	4.6.16
Post test examination	3.39, 3.40 and 3.41	4.6.33
Group 8 (all classes and finishes, except as noted below)		
Visual and mechanical examination	3.1, 3.4, 3.5, 3.6, 3.39 and 3.41	4.5.1
Ice resistance	3.35	4.6.28
Dust (fine sand)	3.36	4.6.29
Impact (classes J and M)	3.48	4.6.40
Post test examination	3.39, 3.40 and 3.41	4.6.33
Group 9 (all classes and finishes, except as noted below)		
Visual and mechanical examination	3.1, 3.4, 3.5, 3.6, 3.39 and 3.41	4.5.1
Cavity-to-cavity leakage bonding integrity 6/	3.37	4.6.30
Insert bond strength 6/	3.46	4.6.38
Thermal vacuum outgassing (classes G and H) 2/ 6/	3.47	4.6.39
Post test examination	3.39, 3.40 and 3.41	4.6.33

See footnotes at end of table.

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TABLE VII. Qualification and qualification retention tests – Continued.

Inspection	Requirement paragraph	Test paragraph
Group 10 (classes J and M)		
Visual and mechanical examination	3.1, 3.4, 3.5, 3.6, 3.39 and 3.41	4.5.1
Hydrolytic stability (classes J and M)	3.49	4.6.41
Post test examination	3.39, 3.40 and 3.41	4.6.33
Group 11 2/, 7/ (all classes and finishes)		
Visual and mechanical examination	3.1, 3.4, 3.5, 3.6, 3.39 and 3.41	4.5.1
Coupling torque	3.10	4.6.3
Backshell shield braid-to-shell conductivity	3.24	4.6.17
Resistance to indirect lightning strike	3.38	4.6.31
Backshell shield braid-to-shell conductivity	3.24	4.6,17
Coupling torque	3.10	4.6.3
Insulation resistance	3.13.1	4.6.6.1
Dielectric withstanding voltage	3.14	4.6.7
Post test examination	3.39, 3.40 and 3.41	4.6.33
Group 12 Electrical (all classes and finishes) 1/, 3/		
Insertion loss	3.3.4.2	4.3.5.1
Return loss	3.3.4.3	4.3.5.2
Near-end cross talk	3.3.4.4	4.3.5.3
Far-end cross talk	3.3.4.5	4.3.5.4
Propagation delay	3.3.4.6	4.3.5.5
Propagation delay skew	3.3.4.7	4.3.5.6

See footnotes at end of table.

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1/ Test groups performed by the qualifying activity as part of retention of qualification.
2/ Initial qualification only.
3/ Test groups performed by the qualifying activity as part of initial qualification. Please note: The qualifying activity does not perform the maintenance aging, altitude low-temperature, shock, and humidity tests.
4/ For retention of qualification testing, random vibration at elevated temperature only (see 4.6.14.1.2 is required).
5/ Insert arrangements using multiaxial contacts (i.e., quadrax or coax contacts) shall not be used for class K and S connectors, as they may fail the firewall test.
6/ Additional parts may be used.
7/ If the supplier's plating system changes (e.g., change to method of plating, or change in plating thicknesses of underplates and finishes), the manufacturer shall notify the qualifying activity and submit a requalification test plan for approval.

4.3.1 Qualification inspection process. At the initiation of the qualification process, prior to testing, prospective suppliers are required to forward a written request for qualification to the qualifying activity (see 6.3). The qualifying activity will respond in writing with a formal "letter of authorization" providing detailed instructions for submission of product samples. Testing cannot begin until the supplier has received this authorization letter. The qualification process is described in the SD-6 (see 6.3.1).

4.3.2 Sample size. The following test samples, manufactured by normal production methods shall be provided for qualification inspection. Tested and untested connectors, accessories, and materials may be required for qualification verification by the responsible qualification activity, at no cost to the Government (see 6.3). Sample size assumes three or more shell sizes are being submitted for qualification. If less than three shell sizes are being submitted for qualification use the small and large shell sizes.

a. Groups 1 and 2. Four mating pairs of straight plugs and wall mounting receptacles of each shell size. Half the samples of each shell size shall have socket contacts in the receptacle and socket contacts in the plug. The other samples shall have socket contacts in the receptacle and pin contacts in the plug. Strain relief clamps, if required to perform testing on qualification samples, shall be in accordance with SAE AS85049.

b. Group 3. One straight plug with spring fingers and counterpart wall mounting receptacle of each shell size unless otherwise specified in the applicable detail specification sheet with applicable EMI backshells. One wall mounting receptacle with applicable finish in each shell size unless otherwise specified in the applicable detail specification sheet, less coupling pins and inserts. The wall mounting receptacle must intermate with the plug.

c. Group 4. Five mated pairs, straight plugs and wall mounting receptacles, of a medium shell size, unless otherwise specified in the applicable specification sheet, in any representative insert and shell size arrangement.

d. Group 5. Five straight plugs and wall mounting receptacles of various representative shell sizes and insert arrangements.

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e. Group 6. Sufficient, mating connectors, plugs, and wall mounting receptacles, in representative shell sizes, to provide a minimum of 21 contact cavities of each applicable contact size.

f. Group 7. For each shell size, one mating pair of connectors, will be supplied with environmental backshells in accordance with SAE AS85049 category 2B. Connectors shall be supplied in maximum density contact arrangements with pin contacts installed in the receptacles.

g. Group 8. One mating pair of straight plug and receptacle of each shell size unless otherwise specified in the applicable detail specification sheet. Each arrangement in a given shell size shall be the most dense contact arrangement.

h. Group 9. Two bonded insert grommet assemblies and two bonded insert interface seal assemblies that have not been assembled into the connector shell shall be tested. The assemblies shall be of the size and contact arrangement for which qualification is to be conducted (the sample size required for thermal vacuum outgassing test shall be as specified in ASTM E595).

i. Group 10. One mating pair of unplated class J or M connectors (less inserts) in each shell size range (small, medium and large) size unless otherwise specified in the applicable specification sheet.

j. Group 11. Three mating pairs of straight plugs and receptacles in shell sizes 9, 15, and 21 (nine mating pairs total), unless otherwise specified in the applicable detail specification sheet, in each class for which qualification is desired shall be tested. See detail specification sheet if connector not specified in shell sizes 9, 15 and 21. Each mating pair of connectors shall be provided with an SAE AS85049 braidsock strain relief as specified in EIA/ECA-364-75. (Note: An SAE AS85049/103 strain relief is recommended.)

k. Group 12. Three mating pairs of straight plugs and receptacles in each shell size, unless otherwise specified in the applicable specification sheet, in each class for which qualification is desired shall be tested.

4.3.3 Preparation of samples. Wiring of crimp-type contacts shall be accomplished using tools conforming to SAE AS22520, class I. The length of stripped portion of conductor shall be at least long enough to reach the bottom of the crimp barrel, but shall not be so long that more than 0.031 inch (0.79 mm) of conductor is exposed at end of barrel when conductor end touches bottom of barrel. An alternate sample preparation may be used with approval of the qualifying activity. Samples provided under 4.3.2.a shall be wired with cables in accordance with SAE AS6070 or equivalent. Specific SAE AS6070 cable specification sheets shall be defined in the applicable MIL-DTL-32546 specification sheet.

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4.3.4 Qualification tests - connector shell and contact.

a. Groups 1 and 2 samples provided under 4.3.2a shall be divided into two similar groups. One group shall be subjected to the tests of group 1, Table VII, in the sequence specified. The other group shall be subjected to the tests of group 2, Table VII, in the sequence specified.

b. Group 3 samples provided under 4.3.2b shall be subjected to the test of group 3, Table VII, in the sequence specified.

c. Group 4 samples provided under 4.3.2c shall be subjected to the tests of group 4, Table VII, in the sequence specified.

d. Group 5 samples provided under 4.3.2d shall be subjected to the tests of group 5, Table VII, in the sequence specified.

e. Group 6 samples provided under 4.3.2e shall be subjected to the tests of group 6, Table VII, in the sequence specified.

f. Group 7 samples provided under 4.3.2f shall be subjected to the tests of group 7, Table VII, in the sequence specified.

g. Group 8 samples provided under 4.3.2g shall be subjected to the tests of group 8, Table VII, in the sequence specified.

h. Group 9 samples provided under 4.3.2h shall be subjected to the tests of group 9, Table VII, in the sequence specified.

i. Group 10 samples provided under 4.3.2i shall be subjected to the tests of group 10, Table VII, in the sequence specified.

j. Group 11 samples provided under 4.3.2j shall be subjected to the tests of group 11, Table VII, in the sequence specified.

k. Group 12 samples provided under 4.3.2k shall be subjected to the tests of group 12, Table VII in the sequence specified.

4.3.5 Qualification tests – electrical. Samples provided under 4.3.2a shall be subjected to the tests of group 12, Table VII in the sequence specified. Refer to TIA-568-C.2, Connecting Hardware Transmission Performance Section for specific measurement values for each test over the frequency described in each detail specification sheet (see 3.50 and Table VII).

a. Measurement shall be performed on a vector network analyzer utilizing 100 ohms (Ω) differential balun fixtures as defined in TIA-568-balun requirements. The common mode balun port termination shall be 50 Ω ±1 percent.

b. Differential mode plus common mode (DMCM) and differential mode (DM) terminations shall be configured as defined in TIA-568-resistance terminations.

c. The maximum length of twisted pair 100 Ω test leads shall be 2 inches (51 mm).

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4.3.5.1 Insertion loss. Insertion loss shall be measured in accordance with ECIA EIA-364-101, Method A. DMCM terminations shall be used on all inactive pairs during test. Measurements shall be performed in at least one plug direction. (See 3.3.4.2).

4.3.5.2 Return loss. Return loss shall be measured in accordance with ECIA EIA-364-108, Test Procedure 4.2. DMCM terminations shall be used on all pairs during test. Measurements shall be performed on both forward and reverse plug directions. (See 3.3.4.3).

4.3.5.3 Near-end cross talk (NEXT). NEXT shall be measured in accordance with ECIA EIA-364-90, Method B. DMCM near end terminations shall be used on adjacent/opposite inactive pairs during test. DMCM far end terminations shall be used on all pairs during test. Measurements shall be performed on all 12 pair measurement combinations in both the forward and reverse plug directions. (See 3.3.4.4).

4.3.5.4 Far-end cross talk (FEXT). FEXT shall be measured in accordance with ECIA EIA-364-90, Method B. DMCM near end terminations shall be used on all inactive pairs during test. DMCM far end terminations shall be used on all inactive pairs during test. Measurements shall be performed on all 6 pair measurement combinations in at least one plug direction. (See 3.3.4.5).

4.3.5.5 Propagation delay. Propagation delay shall be measured in accordance with ECIA EIA-364-103, Method B. DMCM terminations shall be used on all inactive pairs during test. Measurements shall be performed on each pair in at least one direction. (See 3.3.4.6).

4.3.5.6 Propagation delay skew. Propagation delay skew shall be calculated from propagation delay measurements on each of the 6 pair measurement combinations. (See 3.3.4.7).

4.3.6 Failures. One or more failures shall be cause for refusal to grant qualification.

4.3.7 Retention of qualification. Retention of qualification inspection shall occur every 36 months after the initial qualification date. The qualifying activity shall perform or witness the particular tests as specified in Table VII. Unless otherwise specified in the Table VII, sequential testing is required for each group. The qualifying activity may shift the reporting date to accommodate testing schedules, but no later than 18 months from the retention of qualification due date.

4.3.7.1 Samples for the retention of qualification tests. Connector sample units shall be selected and tested in accordance with the following:

a. Groups 1 and 2, test sample quantity and configuration. Two mating plugs and receptacles from each shell size and having each service rating represented shall be provided. One mating pair from each shell size shall be fully wired with the applicable wire listed in Table II maximum diameter and subjected to the applicable tests in Table VII, group 1. The remaining samples shall be fully wired with the applicable wire listed in Table II minimum diameter and subjected to the applicable tests in Table VII, group 2. Vibration testing shall be conducted every three retention of qualification cycles (nine years).

b. Corrosion. The test shall be performed on one receptacle shell and plug shell (with coupling ring) of each finish style qualified. Salt spray shall be performed in accordance with 4.6.9. The standard test (4.6.9.1) is applicable to classes F, G, N and R. The dynamic test (4.6.9.2) is applicable to classes C, H, J, K, L, M, P, S, T, W, X, Y and Z.

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c. Group 1 test sample quantity and configuration. Two mating plugs and receptacles shall be provided for the test. One mating sample of each configuration shall be fully wired in accordance with 4.3.3 and subjected to the applicable tests in Table VII group 1.

d. Group 9, test sample quantity and configuration. Two bonded insert grommet subassemblies and two bonded insert interfacial seal subassemblies with the highest density insert arrangement for each shell size shall be provided for testing. One set of each type of insert assembly shall be installed in the connector shell. Assemblies shall be subjected to the tests in Table VII, group 9, with the exception of thermal vacuum outgassing. The set of inserts assembled into shells shall be subjected to cavity-to-cavity leakage bonding integrity as specified in Table VII, group 9. The set of inserts provided without shells shall be subjected to insert bond strength as specified in Table VII, group 9.

e. Humidity. The humidity test shall be performed on two mating connectors in the largest shell size in regular production during that period (see 4.6.18).

f. High temperature. The high temperature test shall be performed on one mating pair of connectors of any representative shell size for each contact size qualified and arrangement the high temperature exposure test with contact loading shall be in accordance with 4.6.26.1, except that the time period shall be 250 hours.

g. Group 12, test sample quantity and configuration. Two mating plugs and receptacles shall be provided. One mating sample of each configuration shall be fully wired in accordance with 4.3.3 and subjected to the applicable tests in Table VII group 12.

4.4 Conformance inspection.

4.4.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A and B inspections.

4.4.1.1 Group A inspection. Group A inspection shall consist of the inspections specified in Table VIII.

4.4.1.1.1 Visual examination (group A inspection). Each connector and accessory shall be visually examined for completeness, workmanship and identification requirements. Each connector shall be visually inspected for the presence of filler compound in the cavity between wire termination end of insert and shell.

4.4.1.1.2 Critical examinations (group A inspection). The following dimensions and features are considered critical:

a. Mating diameters H and W.

b. Key or keyway positions.

c. Registration of grommet and insert markings (hole pattern between the grommet and the front face of the insert).

d. Red band (mating indicator) in accordance with applicable receptacle detail specification sheet.

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TABLE VIII. Group A inspections. 1/

Inspection	Requirement paragraph	Test paragraph
Visual inspection 2/ 3/	3.1, 3.4, 3.5, 3.6, 3.39 and 3.41	4.5.1
Critical examination 2/ 3/	3.1	4.4.1.1.2
Insulation resistance at ambient temperature 2/ 3/ 4/	3.13.1	4.6.6.1
Dielectric withstanding voltage at sea level 2/ 3/ 4/	3.14	4.6.7.1

1/ Contacts shipped with connectors (except hermetics) shall be from lots that have met the requirements of SAE AS39029.
2/ 100-percent inspection.
3/ The contractor may use in-process controls for this requirement.
4/ Test between two adjacent contacts and between two peripheral contacts and the shell.

4.4.1.2 Group B inspection. Group B inspection shall consist of the inspections specified in Table IX in the order shown and shall be made on sample units that have been subjected to and have passed the group A inspection.

TABLE IX. Group B inspections.

Inspection	Requirement paragraph	Test paragraph
Plating adhesion (classes J and M) 1/ 2/ 3/	3.9	4.6.34
Shell spring finger forces (plugs with spring fingers only) 1/, 2/ 3/	3.26	4.6.19

1/ The contractor may use in-process controls for this requirement.
2/ No failures permitted.
3/ Test 5 pieces.

4.4.1.2.1 Sampling plan (group B inspection). Samples shall be randomly selected as specified in Table X except sampling for plating adhesion and shell spring finger forces shall be as specified in Table V. If one or more defects are found, then the lot shall be rescreened and defects removed. New samples as specified in Table X shall then be randomly selected. If one or more defects are found in the second sample, then the lot shall not be supplied to this specification.

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TABLE X. Group B sampling plan.

Lot size	Sample size
1 to 13	100 percent
14 to 150	13 units
151 to 280	20 units
281 to 500	29 units
501 to 1,200	34 units
1,201 to 3,200	42 units

4.4.1.2.2 Lot definition (group B inspection). The production lot definition is further defined as follows:

a. A production lot consists of all connectors covered under one military specification, manufactured from the specified raw materials, processed under the same specification and procedures, produced by the same type of equipment, and submitted for inspection at one time. Each production lot of assembled connectors shall be a group identified by a common manufacturing record through all significant assembly operations.

b. Common manufacturing records and traceability shall begin with machining and molding and shall include connector assembly. Traceability of connectors to specific physical/chemical test reports of incoming raw material is not required.

4.4.1.2.3 Disposition of sample units. Sample units that have completely passed group B inspection may be delivered on the contract or purchase order if the lot is accepted and sample units are still within specified tolerances.

4.5 Methods of inspection.

4.5.1 Visual and mechanical examination.

4.5.1.1 Connectors. The connectors, accessories, and piece parts shall be visually and mechanically examined to ensure conformance with this specification (see 3.1, 3.4, 3.5, 3.6, 3.39, and 3.41). The contractor may use in-process controls to satisfy these requirements.

4.6 Test methods. The following tests and methods assure connector integrity within typical operating conditions and applications.

4.6.1 Maintenance aging. Connectors shall be tested in accordance with ECIA EIA-364-24. The following detail shall apply: Installing/removal tool shall be in accordance with SAE AS81969/8, SAE AS81969/14 or SAE AS81969/16. A minimum of 20 percent, but not less than four contacts of each connector shall be tested. There shall be no lubricant used during this test. (See 3.7).

4.6.2 Temperature cycling. Mated connectors shall be tested in accordance with ECIA EIA-364-32, test condition A, except that steps 2 and 4 shall be of 2 minutes maximum duration. The temperature during step 1 shall be -65 °C

+0 °C, -5 °C with a high limit of +175 °C. (See 3.8).

4.6.3 Coupling torque. Mated connectors shall be tested in accordance with ECIA EIA-364-114. The following details shall apply (see 3.10):

a. Rate of mating and unmating shall be one inch-pound per second.

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b. There shall be 10 cycles of mating and unmating. Torque shall be measured on the first and last cycle.

c. Torque requirements for mated connectors and protective covers and dummy stowage receptacles shall be as specified in Table IV.

4.6.4 Durability. The wired, assembled plugs, and receptacles shall be mated and unmated 1,500 cycles for classes J and M (contacts may be changed after 500 cycles when contact styles P or S are used), and 500 cycles for all other connectors, at a rate of 300 cycles per hour maximum. The mating and unmating shall be accomplished so that the plug and receptacle are completely separated during each cycle. (See 3.11).

4.6.5 Altitude immersion. Mated connectors shall be tested in accordance with test procedure ECIA EIA-364-03. The following details shall apply (See 3.12):

a. All wire ends shall be located within the chamber and exposed to the chamber atmosphere but not submerged or sealed.

b. At the end of the third cycle, while the connectors are still submerged in the solution, the insulation resistance at ambient temperature shall be measured as required in 4.6.6.1 and the dielectric withstanding voltage test at sea level shall be performed as specified in 4.6.7.1.

4.6.6 Insulation resistance (see 3.13).

4.6.6.1 Insulation resistance at ambient temperature. Unmated connectors shall be tested in accordance with ECIA EIA-364-21. The following details and exceptions apply (see 3.13.1):

a. For lot acceptance testing, where it is undesirable to install actual contacts in connectors, simulated contacts and special techniques may be used in performing this test.

b. The tolerance on the applied voltage shall be ±10 percent.

c. Connectors shall be mated when testing after altitude immersion, humidity and altitude low temperature.

4.6.6.2 Insulation resistance at elevated temperature. . Unmated connectors shall be tested in accordance with ECIA EIA-364-21. The following details shall apply (see 3.13.2):

a. The tolerance on the applied voltage shall be ±10 percent.

b. Elevated temperature is 175 °C +5 °C, -0 °C.

4.6.7 Dielectric withstanding voltage (see 3.14).

4.6.7.1 Dielectric withstanding voltage at sea level. Wired, unmated connectors shall be tested in accordance with ECIA EIA-364-20, method A. Connectors shall be mated when testing after altitude immersion, humidity, and altitude low temperature. The following details and exceptions apply:

a. The magnitude of the voltage shall be as specified in Table I.

b. Fifty percent of the contacts available shall be tested, but in no case shall less than six dielectric withstanding voltage readings be taken. If the number of contacts is four or less, all contacts shall be tested. The test voltage shall be applied between each wired contact and each adjacent contact, and the shell.

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c. The test voltage shall be maintained at the specified value for not less than 2 seconds.

d. For conformance testing, simulated contacts and special techniques may be used in performing the test.

4.6.7.2 Dielectric withstanding voltage at altitude. Mated connectors and unmated connector halves with pin contacts shall be tested in accordance with ECIA EIA-364-20, method A, with the following details and exceptions:

a. The magnitude of the voltage shall be as specified in Table I.

b. Fifty percent of the contacts available shall be tested, but in no case shall less than six dielectric withstanding voltage readings be taken. If the number of contacts is four or less, all contacts shall be tested. The test voltage shall be applied between each wired contact and each adjacent contact and the shell.

c. The test voltage shall be maintained at the specified value for not less than 2 seconds.

d. The leads of all test circuits shall be brought out through the walls of the chamber. There shall be no wire splices inside the chamber. The wire ends of all leads shall be unsealed.

e. The chamber shall be evacuated to each of the specified altitude pressure equivalents listed below:

Altitude	Equivalent pressure
50,000 feet	87 torr
70,000 feet	33 torr

4.6.8 Insert retention. Unmated connectors shall be tested in accordance with ECIA EIA-364-35 with the following details and exceptions (see 3.15):

a. Force to be applied: 100 ±5 pounds per square inch with a 25 pounds minimum force.

b. Connectors may be wired.

4.6.9 Salt spray (corrosion) (see 3.16).

4.6.9.1 Standard test. Unmated connectors shall be tested in accordance with ECIA EIA-364-26. The following details apply.

a. Test condition B (48 hours).

b. The samples shall not be mounted but shall be suspended from the top of the chamber using waxed twine or string, glass rods, or glass cords.

c. Wire ends shall be protected to prevent salt migration.

4.6.9.2 Dynamic test. The wired, assembled plugs and receptacles shall be mated and unmated 50 cycles at a rate of 300 cycles per hour maximum. The mating and unmating shall be accomplished so that the plug and receptacle are completely separated during each cycle. The connectors shall then be subjected to the salt spray test in accordance with ECIA EIA-364-26. The following details and exceptions apply:

a. The connectors shall be tested for 452 hours mated followed by 48 hours unmated.

b. The connectors shall not be mounted, but shall be suspended from the top of the

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chamber using waxed twine or string, glass rods, or glass cord.

c. Wire ends shall be protected to prevent salt migration.

d. After salt spray exposure, the remaining number of durability cycles specified in 4.6.4 shall be completed.

4.6.10 Low signal level contact resistance. The low signal level contact resistance of mated contact pairs shall be measured in accordance with ECIA EIA-364-23 at 25 °C ±3 °C with the connectors fully mated (see 3.17).

4.6.11 Electrical engagement. Counterpart plugs and receptacles shall be wired to provide a complete series circuit through all contacts of the mated connector. A suitable power source and indicator shall be provided such that the earliest point, at which the circuit is completed, during normal connector mating, can be established. Connector halves shall be slowly mated by the normal mating means until first indication of a completed circuit is observed. Means shall be provided to assure that electrical contact is made between the pin and spring of the socket contact, rather than the foot of the socket contact. The mating operation shall be held at this point and the overall connector length shall be measured from solid reference points on the connector halves. The mating operation shall then be continued until the connector halves are in the completely mated position. A second overall length measurement shall then be taken from the same reference points. The difference of these two measurements shall be no less than specified in 3.18.

4.6.12 Contact retention. Unless otherwise specified, contacts shall be tested to ECIA EIA-364-29, procedures I and II, as specified below (see 3.19).

4.6.12.1 Procedure I. Contact retention shall be tested in accordance with ECIA EIA-364-29. The following details and exceptions shall apply:

a. Number of samples – The test shall be performed on 20 percent of the contact complement; but not less than four contacts in each connector half.

b. Applied axial load – Preload to 3 pounds maximum (13.6 Newtons). Apply load as specified in Table XI.

c. Special requirements – Where the test sequence required maintenance aging prior to contact retention, the contacts that were subjected to maintenance aging shall also be selected for contact retention.

d. Axial direction – The applicable forces shall be applied along the longitudinal axis of individual contacts in the direction tending to displace the contacts to the rear.

e. Only the contacts to be tested need be installed in the connector.

4.6.12.2 Procedure II. Contact retention shall be tested in accordance with ECIA EIA-364-29. The following details and exceptions shall apply:

a. 4.6.6.2a through 4.6.6.2b applies.

b. Axial direction – same as 4.6.12.1d, except the direction shall tend to displace the contacts to the front.

c. Only the contacts to be tested need to be installed in the connector.

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TABLE XI. Contact retention.

Contact	Load ±10%	
	pounds	Newtons
22D, 23-22	10	44

4.6.13 Altitude-low temperature. Wired, mated, assembled connectors shall be tested in accordance with ECIA EIA-364-105. The following details apply (see 3.20):

a. No wire ends or splices inside the chamber.

b. Chamber pressure shall be reduced to simulate an altitude of 100,000 feet. The chamber internal temperature shall be reduced to -65ºC, +0ºC, -4ºC. Temperature shall be maintained until the mated pair stabilizes.

c. Insulation resistance test to be performed at 100,000 feet at -65 °C, +0ºC, -4ºC, and shall be in accordance with 4.6.6.1.

d. Dielectric withstanding voltage test to be performed after return to ambient conditions shall be in accordance with 4.6.7.1.

4.6.14 Vibration. Wired and mated connectors shall be subjected to the applicable test(s) specified. Connectors shall be mounted on the vibration table by normal means. All contacts shall carry a test current of 100 milliamperes maximum and shall be continuously monitored for discontinuities throughout the test. A detector capable of detecting any discontinuities in excess of 1 microsecond shall be used. Connectors with accessory threads shall have an accessory load fixture attached during vibration. (See 3.21).

4.6.14.1 Vibration. Connector samples shall be divided equally and subjected to vibration as specified. One connector group shall be subjected to sine vibration in accordance with 4.6.14.1.1, one connector group shall be subjected to random vibration in accordance with 4.6.14.1.2 and the third connector group shall be subjected to random vibration in accordance with 4.6.14.1.3.

4.6.14.1.1 Sine vibration. Connector samples shall be subjected to a simple harmonic motion for 10 to 2,000 Hz in each of three mutually perpendicular axes. The level of vibration shall be a velocity of 254 mm/sec from 10-50 Hz; 1.5 mm double amplitude from 50–140 Hz, and 60 G from 140–2,000 Hz. The entire frequency range from 10-2,000 Hz and back shall be traversed in 20 minutes. The vibration shall be applied for a duration of 12 hours in each of the three mutually perpendicular axes for a total of 36 hours. Each axis of vibration shall be accomplished by vibrating for 4 hours at ambient room temperature, 4 hours at -55° ±5 °C and 4 hours at +175 °C.

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4.6.14.1.2 Random vibration at elevated temperature. Connectors shall be subjected to the test specified in ECIA EIA-364-28. The following details shall apply:

a. Test condition VI – Letter "J" at elevated temperature. The verification of plug to receptacle metal-to-metal bottoming shall be accomplished by demonstrating that the connector meets a minimum rigidity requirement of 300 Hz (i.e. no resonances at frequencies less than 300 Hz). Two accelerometers, one for the perpendicular axis and one for the longitudinal axis shall be used as shown on figure 1. Additionally, an input or control accelerometer shall be placed on non-resonant point of the fixture to control the input vibration. If the accelerometers on the backshell indicate a resonance at frequency below 300 Hz during any part of the vibration test procedures, the connector shall be considered to have failed. Verification of resonance shall be accomplished by monitoring the accelerometers placed on the backshell assembly. Prior to the random vibration test, a 2 g sinusoidal resonance search shall be performed to verify system response. During the random vibration test, the accelerometers data shall be recorded at 1 hour intervals. Following the random vibration test, the sinusoidal resonance search shall be performed again to verify any change in system resonance. In the case of the sinusoidal test, resonance shall be defined as a response greater than twice the input vibration level. In the case of random vibration test, resonance shall be defined as response greater than four times the input vibration level ($1g^2$ vs. $4g^2$). Note: Lightweight accelerometers and mounting features shall be selected so that the total mass of the backshell/cable assembly shall be as specified on figure 2.

b. Vibration to be at 175 °C ±5 °C

c. Duration shall be 8 hours in the longitudinal direction and 8 hours in a perpendicular direction for a total of 16 hours.

4. 6.14.1.3 Random vibration at ambient temperature. Connectors shall be tested in accordance with ECIA EIA-364-28. The following details shall apply:

a. Test condition V – Using the vibration envelope shown on figure 3.

b. Vibration to be conducted at ambient temperature.

c. Duration shall be 8 hours in the longitudinal direction and 8 hours in a perpendicular direction for a total of 16 hours.

4.6.15 Shock. Wired and mated connectors shall be subjected to the applicable test specified in 4.6.15.1 or 4.6.15.2. Connectors shall be mounted by normal means and held together by normal coupling means. All contacts shall be wired in a series circuit with 100 milliamperes maximum current flow through the series circuit during the shock test. Connectors shall be monitored for any discontinuities. A detector capable of detecting all discontinuities of 1 microsecond or longer shall be used. Discontinuity monitoring shall be performed in accordance with ECIA EIA-364-46 (see 3.22).

4.6.15.1 Standard shock. Connectors shall be tested in accordance with ECIA EIA-364-27. The following details shall apply:

a. The pulse shall be an approximated half sine wave of 300 G ±15 percent magnitude with a duration of 3 ±1 milliseconds.

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b. The wire bundle shall be clamped to fixed points at least 8 inches (203.2 mm) from the rear of the connector.

4.6.15.2 High-impact shock. Wired and mated connectors shall be tested in accordance with MIL-DTL-901, grade A with following modification and additions. Mounting fixture shall be in accordance with MIL-DTL-901, light weight. The wire bundle shall be supported on a stationary frame in such a manner as to provide a free flexing length between the frame and fixture of not less than 36 inches (914.4 mm). Each plug shall be terminated with at least 80 percent of wired contacts. The wire bundle shall be provided with a straight, environmental, backshell, category 2B in accordance with SAE AS85049, the longest length available per shell size.

4.6.16 Shell-to-shell conductivity. Mated connectors shall be tested in accordance with ECIA EIA-364-83. (See 3.23).

4.6.17 Backshell shield braid-to-shell conductivity. Mated connectors shall be tested in accordance with ECIA EIA-364-83, except that measurement shall be made between the plug backshell's shield braid and either the accessory threads or mounting flange of the receptacle (see figure 4 for test setup). A backshell in accordance with SAE AS85049 shall be utilized. (See 3.24).

4.6.18 Humidity. Wired, mated connectors shall be tested in accordance with ECIA EIA-364-31. The following details and exceptions shall apply; (see 3.25):

a. Test method IV

b. The mated connectors shall be mounted in a vertical position

c. Three hours minimum after the start of step 7a during the final cycle and while the connectors are still subjected to high humidity, insulation resistance and dielectric withstanding voltage shall be measured when the chamber temperature reaches 20° ±5 °C and condensation is observed on the connector.

d. For qualification testing, insulation resistance readings shall be made on a minimum of 50 percent of the circuits. Outer circuits shall be measured to the connector shell.

4.6.19 Shell spring finger forces. Plugs shall be completely mated and unmated 10 times with counterpart receptacles less coupling pins (if applicable) and inserts. On the mating of the first cycle, the forces necessary to engage the connectors shall be within the values specified in Table V. On the unmating of the last cycle, the forces necessary to separate the connectors shall be within the value specified in Table V. When measuring the engagement and separation forces, the axial loads shall be applied at a displacement rated of 0.5 inch per minute. (See 3.26).

4.6.20 EMI shielding. Applicable to all classes and finishes except class C and finish C. (See 3.27).

4.6.20.1 EMI shielding, low frequencies. Applicable frequency range is 100 to 1,000 MHz only. The EMI shielding effectiveness of mated connectors with EMI backshells (backshells may be soldered to the connector) shall be measured in a triaxial radio frequency leakage fixture as shown in MIL-DTL-38999 (see EMI shielding, low frequencies paragraph of MIL-DTL-38999). The EMI leakage from the conductor inside the connector in the same inner coaxial line into the outer coaxial line shall be measured at the frequencies specified in MIL-DTL-38999 (see EMI shielding, low frequencies paragraph of MIL-DTL-38999) within a

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frequency accuracy of ±5 percent. The level of detected signal power shall be specified by a tunable radio frequency field intensity meter isolated from the test circuit by a 3 to 10 dB pad. Care shall be taken to ensure that the signal is a result of EMI leakage from within the mated connector and not due to a faulty termination inside the fixture. All terminations inside the fixture, whether to the EMI back shells or between internal conductors, shall have a leakage at least 10 dB less than the test requirements. The signal source shall be set to the desired frequency. The signal shall be fed through a 3 to 10 dB isolation pad to a parallel circuit consisting of a Double Pole Double Throw (DPDT) coaxial switch connected so that the signal can be manually or electronically fed alternatively to the fixture and to a variable 100 dB reference attenuator. The attenuator shall be adjustable in 1 dB steps and calibrated to ±3 dB.

a. The inserts may be removed from the connectors under test or the contacts removed and a hole drilled through the inserts to accommodate a center conductor of suitable geometry to provide a good 50-ohms impedance match with inside diameter of the mated connector shells and without introducing significant discontinuities in the line. The maximum voltage standing ware ratio (VSWR) in the inner coaxial line shall be 1.5. The outer shell of the test fixture shall be so constructed as to provide a good 50-ohms impedance match with the outside diameter of the mated connector shells, coupling ring, and EMI backshells. The maximum VSWR of the outer coaxial line shall be 1.5.

b. A sliding circumferential short shall be positioned behind the connector on the signal input end of the fixture to provide for tuning the outer coaxial line for maximum output at each test frequency. The allowable travel of this short shall be greater than ½ wave lengths at the lowest test frequency of 1.5 meters minimum for 100 MHz The inner coaxial line shall be terminated in fixed 50-ohms load impedance behind the connector at the output end of the fixture.

c. The connector used to couple together the various elements of the test system shall be of a low-leakage type which has a nominal impedance of 50 ohms, a VSWR of less than 1.5, and a minimum leakage attenuation of 100 dB. The output impedance of the signal source and the input impedance of the detector shall be nominally 50 ohms with a maximum VSWR of 1.5. The input and output VSWR of the standard attenuator shall be less than 1.5 in the 20 to 100 range.

d. The relative signal level in the variable attenuation shall be equaled to the signal level through the leakage fixture by adjusting the attenuator. The signal loss in the fixture can then be read from the setting on the variable attenuator.

4.6.20.2 EMI shielding, high frequencies. Applicable frequency range is 1,000 to 10,000 MHz only. The EMI shielding effectiveness of mated connectors with EMI backshells shall be measured using the mode-stirred technique in accordance with ECIA EIA-364-66.

4.6.21 Ozone exposure. Wired, mated connectors shall be tested in accordance with ECIA EIA-364-14. (See 3.28).

4.6.22 Fluid immersion. Connector samples shall be tested in accordance with ECIA EIA-364-10 (one sample per fluid). Following the fluid immersion cycles, the connectors shall be tested for coupling torque in accordance with 4.6.3 and dielectric withstanding voltage at sea level in accordance with 4.6.7.1 within 3 hours. (See 3.29).

4.6.22.1 Retention system fluid immersion. Unmated connectors with contacts removed shall be immersed in the fluids in accordance with ECIA EIA-364-10 (one sample per fluid) for 2

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hours at room temperature. After removal, excess fluid shall drain from the connectors for 4 hours and the contacts reinstalled. Following the test, the connectors shall be subjected to contact retention as specified in 3.19. (See 3.29.1).

4.6.23 Pin contact stability. The unmated connectors shall have 10 percent of their pin contacts subjected to this test. Gauge pins conforming dimensionally to the requirements on figure 5 shall be used. The connector shall be held in a holding device. The forces specified in Table VI shall be applied to the exposed rod as shown on figure 6. The load shall be applied gradually at a rate not exceeding 1 inch (25.4 mm) per minute. The total pin tip displacement shall be measured as shown on figures 6 and 7. (See 3.30).

4.6.24 Contact walkout. Two contacts in each plug and receptacle shall be tested. The contacts shall be crimped to stranded steel cable of an appropriate size and installed in the connector. The unmated connector shall be mounted in a test fixture as shown on figure 8. A 3-pound load shall be applied to the cable. One 360° rotation of the fixture with the connector mounted shall constitute on cycle. The connector shall be subjected to 100 cycles at a rate of 10 to 20 cycles per minute. (See 3.31).

4.6.25 Installing and removal tool abuse (applicable only at initial qualification). Five contact cavities in each connector shall be subjected to the tests specified in 4.6.25.1. Different contact cavities shall be used for each test. If a tool becomes damaged during any of the testing, it shall be replaced. Failure of a tool shall not constitute a test failure. Tools shall be utilized in accordance with SAE AS81969/14 or SAE AS81969/16-04, as applicable (see 3.32).

4.6.25.1 Removal tool rotation. The tool shall be inserted as if to remove a contact and a total of 3 pounds axial load shall be applied. With the force applied, the tool shall be rotated 180° and then removed also removing the contact. The contact shall be reinserted. The steps shall be repeated three times on each of five contacts selected.

4.6.26 High temperature exposure (see 3.33).

4.6.26.1 High temperature exposure with contact loading. Connectors with removable contacts shall have one mating pair of contacts removed from an untested contact cavity and replaced with contacts crimped or otherwise attached to steel cable or steel-cored copper wire (copper-weld, or equivalent) of an appropriate size. The axial location of these contacts shall be measured for conformance to the applicable dimensions on figure 9, as applicable, with a load of approximately 2 pounds applied to seat the contact back against the retention device. The connector shall then be mounted in a fixture as shown on figure 10. A weight equal to 50 percent of the axial load specified in Table XI for the applicable contact size shall be suspended freely from each steel wire. A current of 100 milliamperes maximum shall be applied to the test contacts and a suitable instrument shall be used to monitor the circuit for discontinuities in excess of 1 microsecond. The mounted connector shall then be exposed to 175 °C for classes J, P, T, W, X and Z and 200 °C for all other finishes and classes for 1,000 hours minimum. After the connectors return to ambient temperature, they shall be unmated and the contact locations shall be re-measured with approximately 2 pounds axial load applied to the contact back against the retention device.

4.6.26.2 High temperature exposure. Mated connectors shall be subjected to an ambient temperature of 175 °C +3 °C, -0 °C for classes J, P, T, W, X and Z, and 200 °C + 3 °C, -0 °C for all other classes. The temperature shall be maintained for 1,000 hours.

4.6.27 Electrolytic erosion (applicable to all classes). The wired receptacle half of the

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connector containing the pin contacts shall be firmly mounted in a vertical position pins facing up. A salt-water solution (5 percent salt by weight) shall be poured onto the pin interface until the solution level reaches the pin tips. The solution shall remain in the connector for a minimum of 30 seconds and then poured out, lightly shaking excesses from the connector. The counterpart plug shall immediately be mated with the receptacle. One cell pattern, with all contacts adjacent to a central contact at a positive polarity and center contact at a negative polarity, shall be set up at a location giving the most uniform distribution of contacts adjacent to the center contact. A potential of 60 volts DC (VDC) shall be applied between the contacts adjacent to the center contact. The contacts shall be energized for 40 hours. Following this, with voltage disconnected, the connectors shall be unmated and the pin contacts removed. The contacts shall be examined under three power (3X) magnification for erosion to base metal. (See 3.34).

4.6.28 Ice resistance (applicable to all classes). Mated connectors shall be immersed in tap water for 1 minute and then placed in an ambience of -65 °C for 1 hour. A minimum of three such cycles shall be performed until the connector surfaces are completely iced over. Immediately after removal from the last cycle, the frozen connectors shall be uncoupled and then recoupled. The connectors shall be uncoupled and recoupled a second time with uncoupling and coupling torque measured in accordance with 4.6.3. The uncoupling and coupling torque shall not be measured on the first uncoupling and recoupling. (See 3.35).

4.6.29 Dust (fine sand) (applicable to all classes). Mated connectors shall be tested in accordance with MIL-STD-202-110. (See 3.36).

4.6.30 Cavity-to-cavity leakage bonding integrity. Unmated connectors shall be tested in accordance with ECIA EIA-364-78. (See 3.37).

4.6.31 Resistance to indirect lightning strike. Mated connectors shall be tested in accordance with ECIA EIA-364-75 with the following details and exceptions. (See 3.38).

a. Test sequence shall be as specified in Table VII, group 11.

b. Each specimen shall be subjected to three current pulses in each polarity (a total of six pulses). After each current pulse, the connectors shall be unscrewed ½ turn counterclockwise without removing or unmating in order to break any weld points that may have occurred. Connectors shall then be re-tightened ½ turn, back to the fully mated position.

c. Visual examination shall be conducted under three-five power (3-5X) magnification in accordance with ECIA EIA-364-75.

d. Coupling torque shall be in accordance with 4.6.3.

e. Shell-to-shell conductivity testing shall be in accordance with 4.6.16.

f. The peak current, I_{pk}, shall be as specified in Table XII.

g. Insulation resistance at ambient temperature shall be in accordance with 4.6.6.1, followed by dielectric withstanding voltage at sea level in accordance with 4.6.7.1.

h. The waveform for indirect lightning strike effects, as specified on figure 11, shall be utilized. Indirect lightning strike test setup, as specified on figure 12, shall be utilized.

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TABLE XII. Resistance to indirect lightning strike test currents.

Connector Class	Peak current (amps)
Class C	Not applicable
Classes F, G, H, K, L, P, R, T, S, X, W, Y and Z	10,000
Classes J and M (shell sizes 11 and larger)	6,000
Classes J and M (shell sizes 9)	3,000

4.6.32 Magnetic permeability. Connectors shall be tested in accordance with ECIA EIA-364-54. (See 3.4.3).

4.6.33 Post test examination. The tested connectors and contacts shall be examined for compliance with 3.39, 3.40 and 3.41 to determine the effects of previous testing. Any evidence of cracking, loosening of parts, carbon tracking, excess wear, or missing parts shall be recorded.

4.6.34 Plating adhesion (applicable to all classes) (see 3.9). Immerse plated connector shells in canola, peanut or other similar oil. The oil shall be heated to the maximum operation temperature of the connector (see 1.2.3). Leave the connector shells in the boiling oil for 2 minutes. Then remove the shells and within 10 seconds transfer it to an ambient temperature solvent or ice water. Let stand for 45 seconds. Then remove and examine visually for conformance to the requirements of 3.9. Optional procedure: Place connector shells into a hot oven until shells are heated to the maximum operational temperature of the connector. Once the maximum temperature is attained, the shells shall remain in the oven for an additional 2 minutes, after which they shall be removed and transferred to an ambient temperature solvent or ice water within 10 seconds. Shells shall remain in solvent or ice water for 2 minutes, then removed and examined visually for conformance to the requirements of 3.9.

4.6.35 Gauge location. Applicable test gauges, specified on figures 5, 13, and 14, shall be installed in three randomly selected cavities in each connector, with the accessory rear hardware removed. With each test gauge fully seated back against its contact retention device, the axial location of the front end of each gauge shall be measured relative to the reference point specified on figure 9, as applicable. The test shall be repeated on pin assemblies using gauges specified on figure 13 and seating them forward in the contact cavity. (See 3.43).

4.6.36 Gauge retention. Applicable test gauges, specified on figures 5 and15, shall be installed in three randomly selected cavities in each connector, with the accessory rear hardware removed. The axial load specified in Table XI shall be applied to individual test gauges in both directions. The load shall be applied at a rate of approximately 1 pound per second until the specified load has been reached. Gauge displacement shall be measured with respect to the connector shell after an initial load of 2 pounds has been applied to ensure that all slack has been taken up. (See 3.44).

4.6.37 Fungus resistance certification. Fungus testing shall be conducted in accordance with MIL-STD-810, method 508.6. A certificate signed by an individual authorized to speak for the manufacturer may by submitted in lieu of fungus testing with the approval of the qualifying activity. The certificate shall state that the generator system components are constructed of non-nutrient materials and do not use any organic material. (See 3.45).

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4.6.38 Insert bond strength. The connector's bonded insert assembly shall be tested in accordance with ECIA EIA-364-79. (See 3.46).

4.6.39 Thermal vacuum outgassing (applicable to classes G and H only for initial qualification). All materials used in the finished connector shall be tested in accordance with ASTM E595. Samples to be tested shall have been processed in the same manner as that used in production of the qualification lot (see 3.47).

4.6.40 Impact (applicable only to classes J and M). Connector plugs with SAE AS85049 straight strain relief clamps shall be tested in accordance with ECIA EIA-364-42. The following conditions apply (see 3.48):

a. Drop height shall be 4 feet (1.219 meters).

b. Number of drops shall be 8.

c. Plate shall be indexed at 36° intervals.

d. Plugs shall have no caps or covers installed.

4.6.41 Hydrolytic stability (classes J and M) (initial qualification). Mated connectors less inserts shall be subjected to the following water absorption test (connectors shall be unplated). The connectors shall be tested in accordance with ASTM D570 for long-term immersion (see 3.49).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract of order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or with the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files. CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The connectors qualified to this specification are intended for use as high-speed data connectors using copper conductor for transmission of signals up to 10 gigabits per second (Gbps), such as IEEE 802.3 10GBASE-T, over four-pair cabling with cable lengths up to 328 feet (100 meters). This connector is military unique because it is a miniature, high density, bayonet, threaded, breach coupling, environment resisting, high shock, high vibration, high corrosion connector that uses crimp removable contacts. It is capable of operating at temperatures up to 200 °C. It is also designed to withstand 500 hours of salt spray. A full commercial substitute is not available.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Assembly instructions, if required (see 3.4.2).

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c. Packaging requirements (see 5.1).

d. Nomenclature by class, type, finish, style, size, and PIN (see 1.3, 3.39 and 3.40).

e. Specify if contacts, sealing plugs, sealing boots and /or tools are included (see 1.2.4, 3.3.3, 3.4.5.1, 3.5.2, 3.5.3 and 3.5.6.2).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Product List (QPL) No. 32546 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from the Naval Air Systems Command (Code 4.4.5.3), 48298 Shaw Road, Bldg. 1461, Patuxent River, MD 20670-1900 or QPL_Support@navy.mil. An online database of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.3.1 Provisions governing qualification. Qualification will be conducted in accordance with the provisions and guidelines of SD-6, “Provisions Governing Qualification.” Copies of this document are available online at https://assist.dla.mil.

6.3.2 QPL supplemental information. The qualifying activity provides a summarized list of all qualified sources on a publicly accessible electronic site. The summary includes but is not limited to the supplier approved part number and related specification part number, a dedicated approval reference number, a supplier location where purchases may be requested and the manufacturing location of the component. The suppliers and products qualified to this specification are available on the qualifying activity website at http://www.navair.navy.mil/qpl/.

6.4 Subject term (key word) listing.

Contacts
High temperature
Nominal impedance

6.5 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notation and relationship to the previous issue.

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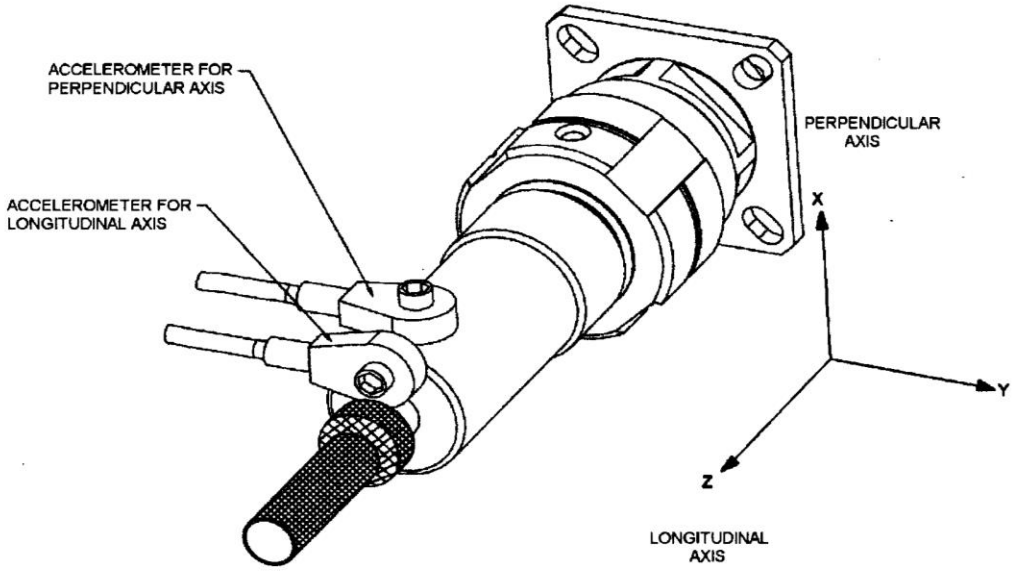
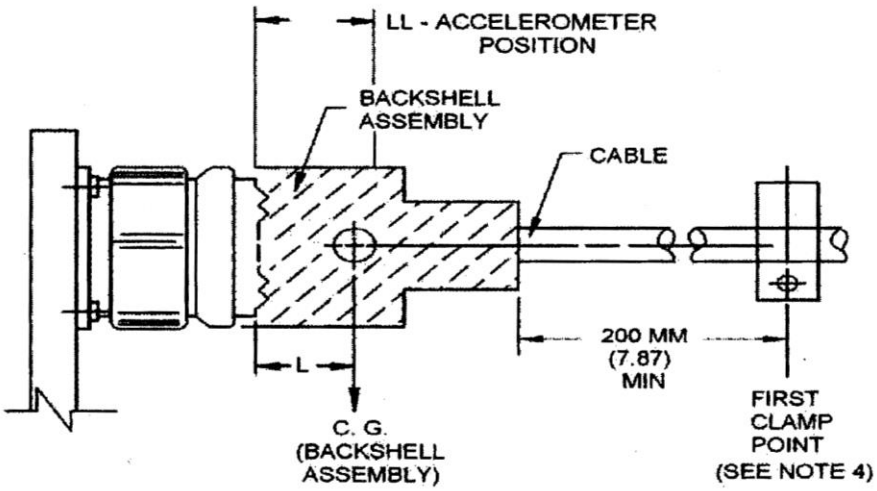


FIGURE 1. Vibration test setup for metal-to-metal shell bottoming verification.

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Shell size	L (mm) root to C.G. + 10% - 0%	Weight (kg) backshell assembly + 10% - 0%	LL (mm) ± 1
9	15.24	.05	10.2
11	17.78	.06	10.2
13	19.05	.07	17.8
15	19.05	.09	25.4
17	19.05	.11	25.4
19	22.86	.18	30.5
21	22.86	.22	30.5
23	23.37	.25	35.6
25	23.37	.31	35.6

NOTES:
1. Dimensions are in millimeters.
2. Shape of test accessory is optional.
3. The test accessory may include a strain relief clamp.
4. For series III, clamp point to be located on vibration table, and cable weight between rear of grommet and tie down clamp point to be considered a portion of the backshell assembly weight. Clamp point for series I and IV is to be on nonvibrating member.

FIGURE 2. Vibration test accessory.

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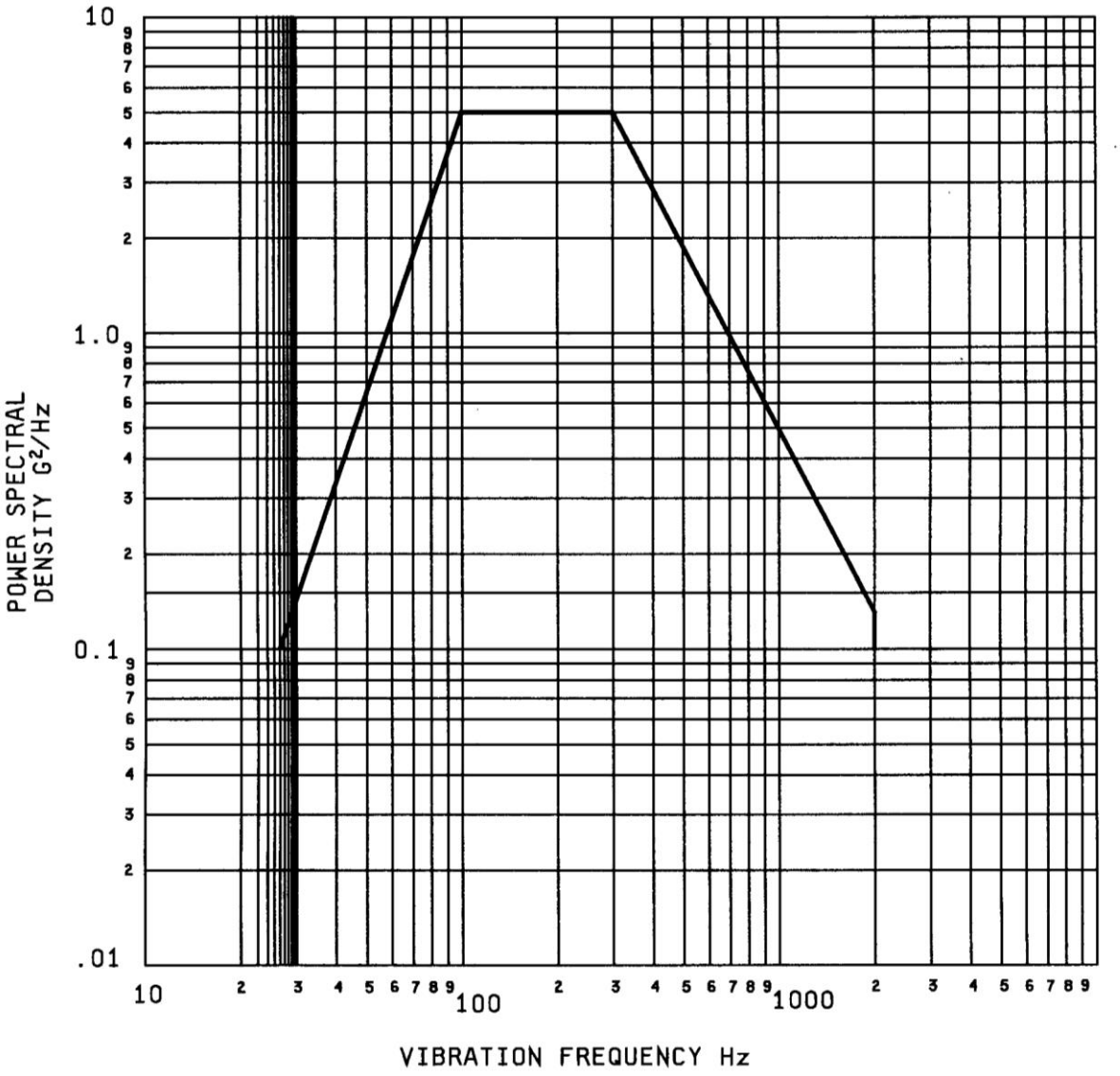


FIGURE 3. Random vibration.

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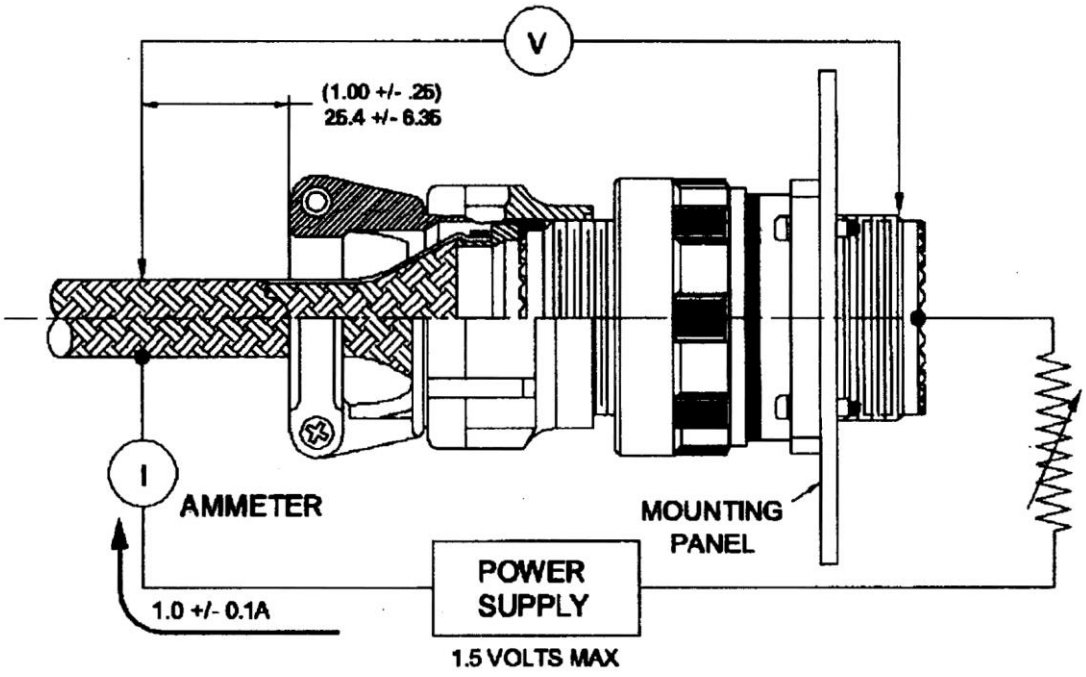


FIGURE 4. Backshield braid-to-shell conductivity.

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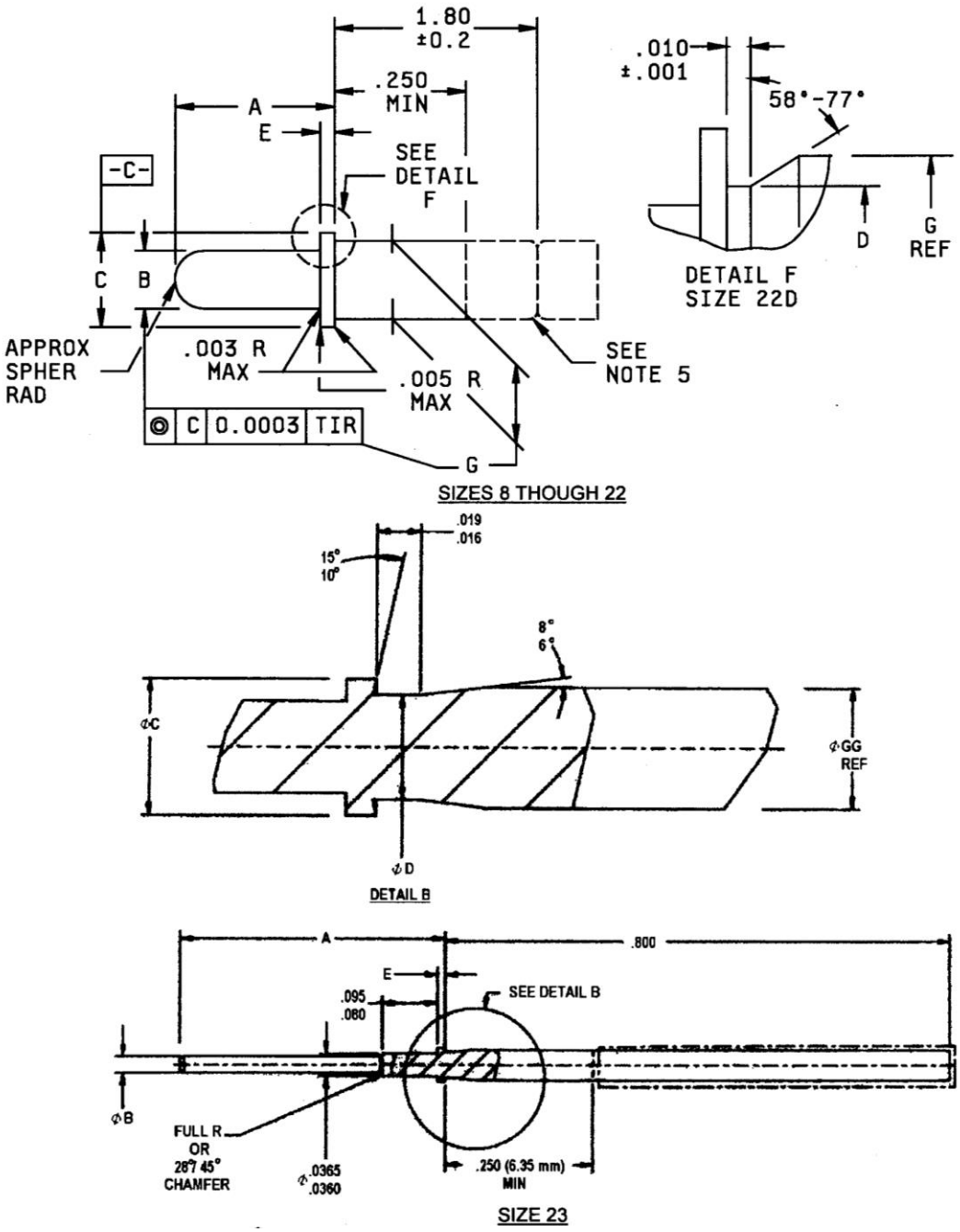


FIGURE 5. Test gauge, pin, gauge location, and retention.

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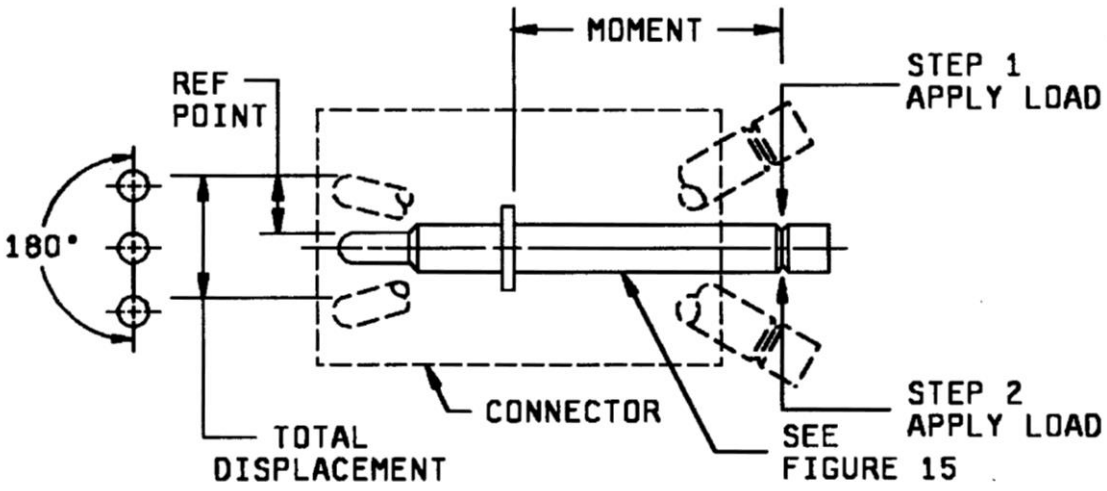
Contact size	A +.0005 -.0000	B dia +.0002 -.0000	C dia +.0002 -.0000	D dia +.0000 -.0002	E +.0000 -.0002	G dia +.0000 -.0002
23	.4200	.0265	.0540	.0415	.0120	.0480
22D	.2950	.0295	.0600	.0410	.0330	.0480
20	"	.0390	.0910	---	"	.0700
16	"	.0615	.1270	---	"	.1030
12	"	.0930	.1790	---	"	.1510
10	"	.1240	.2380	---	"	.2130
8 twinax or coax	"	.2170	.3110	---	"	.2760

Inches	mm	Inches	mm	Inches	mm
.0002	0.005	.0390	0.991	.1270	3.226
.0003	0.008	.0410	1.041	.1510	3.835
.0005	0.013	.0480	1.219	.1790	4.547
.001	0.03	.0600	1.524	.2130	5.41
.003	0.08	.0615	1.562	.2170	5.51
.005	0.13	.0700	1.778	.2380	6.05
.010	0.25	.0910	2.311	.250	6.35
.02	0.5	.0930	2.362	.2760	7.01
.0295	0.749	.1030	2.616	.2950	7.493
.0330	0.838	.1240	3.15	1.80	45.7

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Material: Hardened tool steel.
4. Finish: 32 microinches polished.
5. Design of rear extension is optional, but must have a groove provided as specified.

FIGURE 5. Test gauge, pin, gauge location, and retention - Continued.

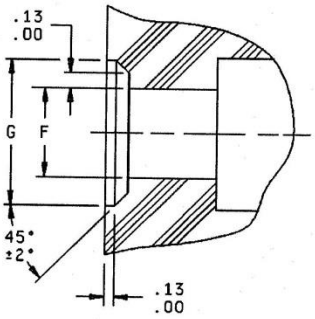
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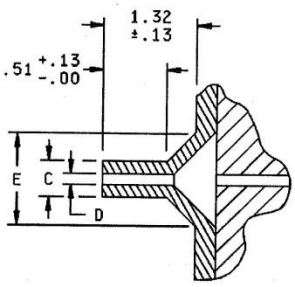
Step 1 - Apply load to determine reference point.
Step 2 - Apply load to opposite direction (180°) and measure total displacement.

FIGURE 6. Pin contact stability test.

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DETAIL A SOCKET ENTRY



DETAIL B PIN BARRIER

Contact size	Detail A		Detail B		
	F dia	G dia	E dia	C dia	D dia max
23	0.89 (.035) 0.81 (.032)	1.52 (.060) 1.42 (.056)	1.50 (.059) 1.37 (.054)	1.22 (.048) 1.14 (.045)	0.64 (.025)
22D	0.97 (.038) 0.89 (.035)	1.70 (.067) 1.60 (.063)	1.88 (.074) 1.75 (.069)	1.35 (.053) 1.27 (.050)	0.71 (.028)
20	1.32 (.052) 1.24 (.049)	2.26 (.089) 2.16 (.085)	2.44 (.096) 2.31 (.091)	1.91 (.075) 1.83 (.072)	0.97 (.039)
16	1.88 (.074) 1.80 (.071)	2.84 (.112) 2.74 (.109)	3.02 (.119) 2.90 (.114)	2.49 (.098) 2.41 (.095)	1.52 (.060)
12	2.69 (.106) 2.62 (.103)	3.66 (.144) 3.56 (.140)	3.84 (.151) 3.71 (.146)	3.30 (.130) 3.23 (.127)	2.34 (.092)
10	3.48 (.137) 3.40 (.134)	4.45 (.175) 4.34 (.171)	4.62 (.182) 4.50 (.177)	4.09 (.161) 4.01 (.158)	3.12 (.123)
8 twinax or coax	5.84 (.230) 5.77 (.227)	6.81 (.268) 6.71 (.264)	7.06 (.278) 6.93 (.273)	6.50 (.256) 6.38 (.251)	5.49 (.216)

NOTES:
1. Dimensions are in millimeters.
2. Inch equivalents (in parentheses) are given for information only.
3. Detail A is not applicable to hermetic or thru-bulkhead receptacles.

FIGURE 7. Interface design.

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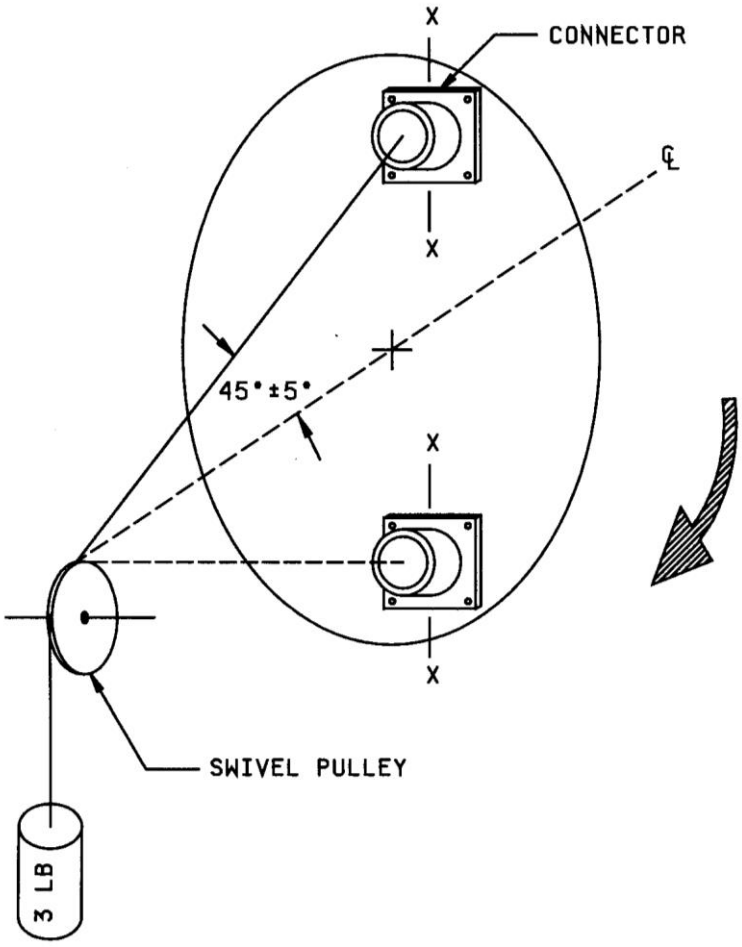


FIGURE 8. Contact walk-out test setup.

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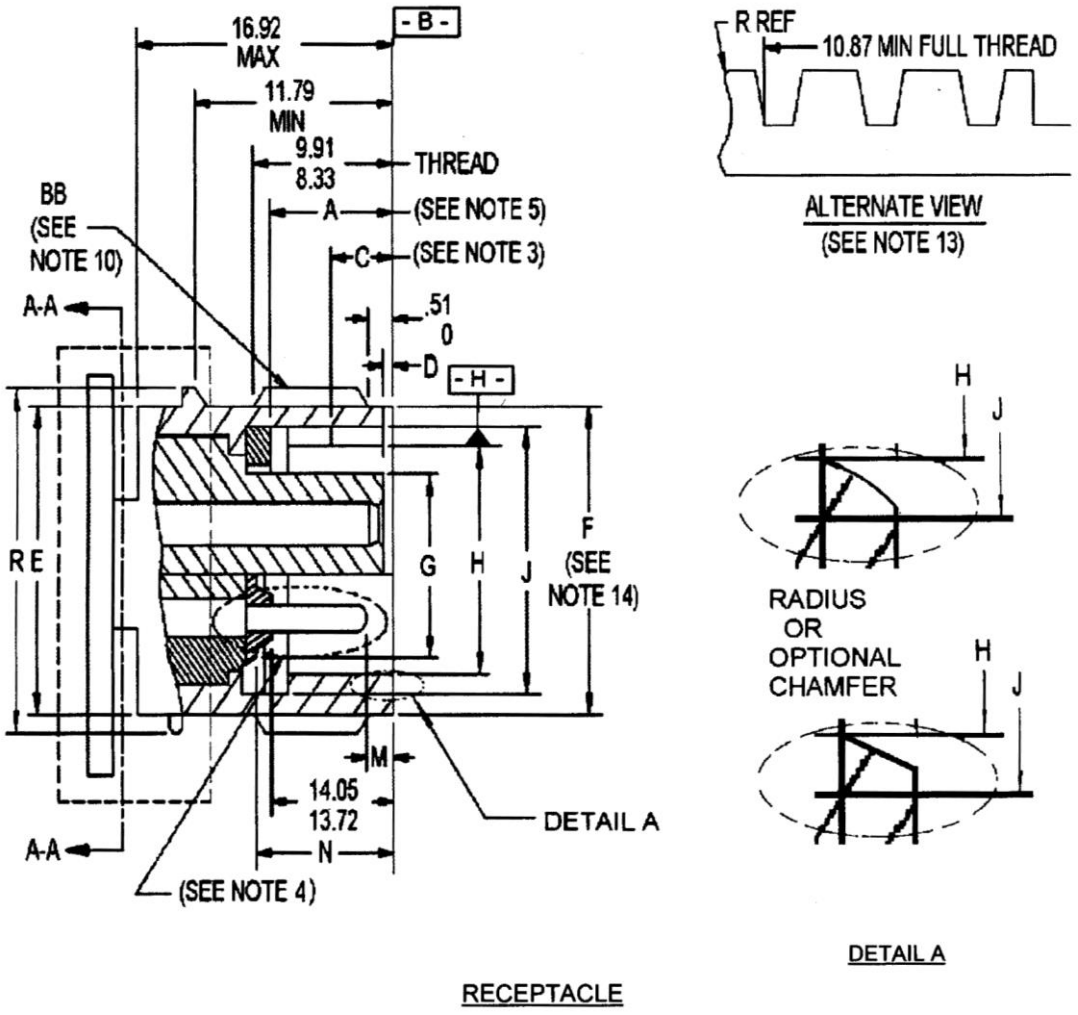


FIGURE 9. Connector intermateability dimensions.

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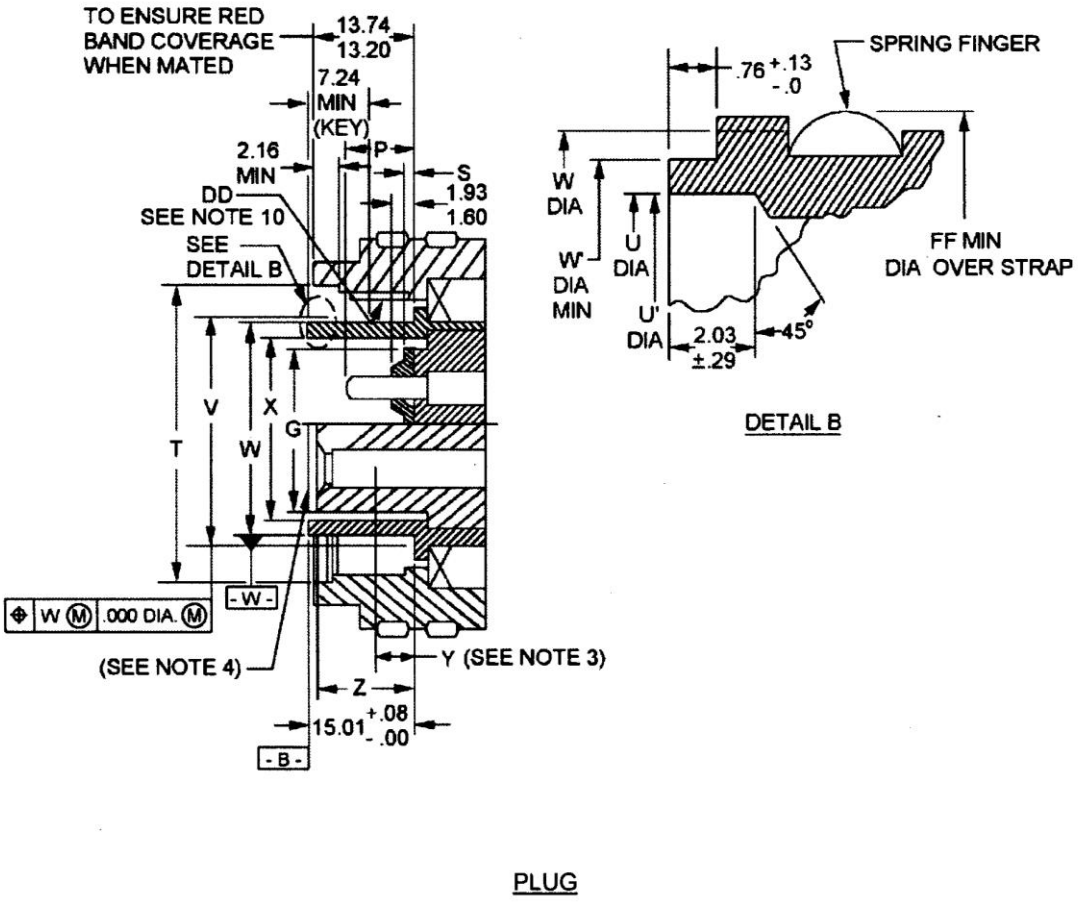


FIGURE 9. Connector intermateability dimensions - Continued.

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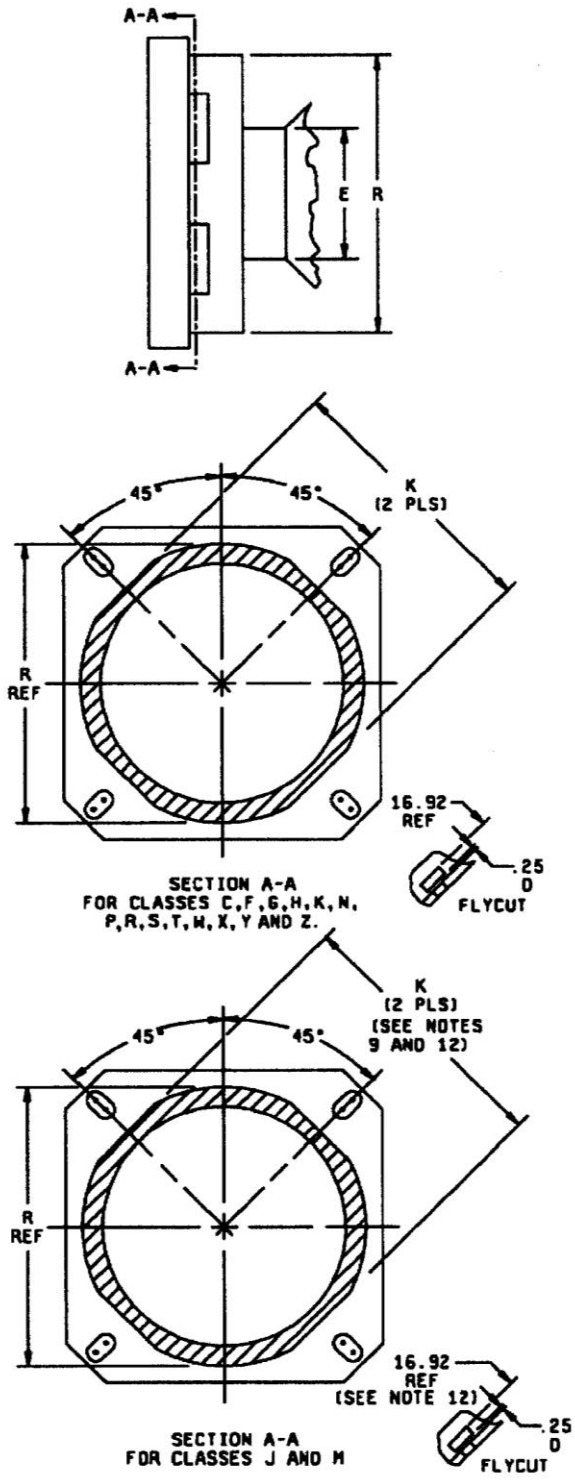


FIGURE 9. Connector intermateability dimensions - Continued.

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Shell size	A Gasket location	C (see note 3)	D Socket insert location	E dia max	F dia
9	14.73 14.50	3.25 max	0.94 0.64	14.55	14.42 14.14
11				17.73	17.60 17.32
13				20.90	20.77 20.49
15				24.08	23.95 23.67
17				28.52	28.30 27.97
19				30.12	29.89 29.56
21				33.30	33.07 32.74
23				36.47	36.24 35.91
25				39.65	39.42 39.02

FIGURE 9. Connector intermateability dimensions - Continued.

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Shell size	FF dia min	G dia +.13 -.25	H dia (except classes J & M)	H dia (classes J & M only)	J dia (except classes J & M)	J dia (classes J & M only)
9	11.48	7.24	11.36 11.20	11.43 11.20	12.63 12.34	12.70 12.34
11	14.73	10.49	14.61 14.45	14.68 14.45	15.88 15.60	15.95 15.60
13	17.58	13.39	17.45 17.30	17.52 17.30	19.66 19.38	19.73 19.38
15	20.75	16.56	20.63 20.44	20.70 20.44	22.84 22.56	22.91 22.56
17	23.92	19.74	23.80 23.64	23.87 23.64	26.01 25.73	26.08 25.73
19	26.59	22.00	26.47 26.31	26.54 26.31	28.63 28.40	28.70 28.40
21	29.77	25.17	29.64 29.49	29.71 29.49	31.85 31.57	31.92 31.57
23	32.94	28.35	32.82 32.66	32.89 32.66	35.03 34.75	35.10 34.75
25	36.12	31.52	35.99 35.84	36.06 35.84	38.20 37.92	38.27 37.92

FIGURE 9. Connector intermateability dimensions - Continued.

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Shell size	K max	M pin contact location	N pin insert location	P pin contact location	R dia max	S pin insert location	T dia +.25 -.03
9	11.84				15.88		16.51
11	15.01				19.05		19.69
13	19.08				22.23		22.86
15	22.25	10.54	15.46	6.15	25.40	0.65	26.04
17	25.43	9.50	15.04	5.11	30.16	0.24	30.89
19	28.60				31.75		32.49
21	31.78				34.73		35.66
23	34.95				38.10		38.84
25	38.13				41.28		42.01

Shell size	U dia min/max	U dia +.13 -.03	U' dia min	V dia over keys	W dia between keys (except classes J & M)	W dia between keys (classes J & M only)	W' dia min front of keys	X dia min	Y (see note 3)	Z socket insert location
9	9.96 9.55	9.83	8.51	12.27 11.99	11.13 10.97	11.13 10.85	10.59	7.92		
11	13.19 12.61	13.06	11.74	15.52 15.24	14.38 14.22	14.38 14.10	13.84	10.92		
13	16.08 15.45	15.95	14.63	19.31 19.02	17.22 17.07	17.22 16.95	16.68	13.81		
15	19.26 18.63	19.13	17.81	22.48 22.20	20.40 20.24	20.40 20.12	19.86	16.99	12.45 min	15.06 14.76
17	22.43 21.80	22.30	20.98	25.66 25.37	23.57 23.42	23.57 23.30	23.03	20.16		
19	24.82 24.47	24.69	23.57	28.32 28.04	26.24 26.09	26.24 25.97	25.70	22.42		
21	27.99 27.64	27.86	26.74	31.50 31.22	29.42 29.26	29.42 29.14	28.87	25.60		
23	31.17 30.82	31.04	29.92	34.67 34.39	32.59 32.44	32.59 32.32	32.05	28.77		
25	34.34 33.99	34.21	33.09	37.85 37.57	35.77 35.61	35.77 35.49	35.22	31.92		

FIGURE 9. Connector intermateability dimensions - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:58Z
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BB thread (plated or finished) (see notes 10 and 17)										
Shell size	Designation				External thread limits of size class 2A					
	Thread size	Pitch	Lead	Allowance (see note 18)	Major diameter Limits tolerance is .0080		Pitch diameter Limits tolerance is .0080		Minor diameter Limits	
					max	min	max	min	max	min
9	.6250	.1	.3	.0015	.6235	.6155	.5975	. 5895	.5675	.5535
11	.7500				.7485	.7405	.7225	. 7145	.6925	.6785
13	.8750				.8735	.8655	.8475	.8395	.8175	.8035
15	1.0000				.9985	.9905	.9725	.9645	.9425	.9285
Shell size	Thread size	Pitch	Lead	Allowance (see note 18)	Major diameter Limits tolerance is .0120		Pitch diameter Limits tolerance is .0100		Minor diameter Limits	
					max	min	max	min	max	min
17	1.1875	.1	.3	.0020	1.1855	1.1735	1.1515	1.1415	1.1135	1.0955
19	1.2500				1.2480	1.2360	1.2140	1.2040	1.1760	1.1580
21	1.3750				1.3730	1.3610	1.3390	1.3290	1.3010	1.2830
23	1.500				1.4980	1.4860	1.4640	1.4540	1.4260	1.4080
25	1.625				1.6230	1.6110	1.5890	1.5790	1.5510	1.5330

FIGURE 9. Connector intermateability dimensions - Continued.

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DD thread (plated) (see notes 10 and 17)										
Shell size	Designation			Internal thread limits of size class 2B						
	Thread size	Pitch	Lead	Minor diameter		Pitch diameter		Major diameter		
				Limits tolerance is .0100		Limits tolerance is .0100		Limits		
				min	max	min	max	min	max	
9	.6250	.1	.3	.5792	.5892	.5990	.6090	.6290	.6450	
11	.7500			.7042	.7142	.7240	.7340	.7540	.7700	
13	.8750			.8292	.8392	.8490	.8590	.8790	.8950	
15	1.0000			.9542	.9642	.9740	.9840	1.0040	1.0200	
Shell size	Thread size	Pitch	Lead	Minor diameter		Pitch diameter		Major diameter		
				Limits tolerance is .0120		Limits tolerance is .0120		Limits		
				min	max	min	max	min	max	
17	1.1875	.1	.3	1.1274	1.1394	1.1535	1.1655	1.1915	1.2115	
19	1.2500			1.1900	1.2020	1.2160	1.2280	1.2540	1.2740	
21	1.3750			1.3150	1.3270	1.3410	1.3530	1.3790	1.3990	
23	1.5000			1.4400	1.4520	1.4660	1.4780	1.5040	1.5240	
25	1.6250			1.5650	1.5770	1.5910	1.6030	1.6290	1.6490	

FIGURE 9. Connector intermateability dimensions - Continued.

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NOTES:

1. Dimensions are in millimeters (except coupling threads).
2. Inch equivalents are given for general information only.
3. A point at which a gauge pin, having the same basic diameter as the mating contact and a square face, touches socket contact spring.
4. Details "A" (socket entry) and "B" (pin barrier) on figure 14 apply to both plugs and receptacles.
5. "A" initial contact with static seal.
6. Insert front surface shall be flat within 0.13 mm TIR.
7. Diameters "H" and "G" shall be concentric within 0.13 mm TIR at MMC.
8. Diameter "W" with respect to diameter V shall be concentric at MMC.
9. Dimension "K" is for clearance of mounting screw heads.
10. Thread approach modified to eliminate cross-threading.
11. Receptacle coupling threads may have modified minor diameters.
12. Flycuts are an optional design for composite.
13. The "E" dimension does not apply.
14. Witness marks as result of threading are permissible on "F" diameter.
15. Diameter "W" (dia between keys) is for metallic and composite plugs, diameter "W' (W prime min – dia front of keys)" is for metallic and composite plugs, diameter "U" is for metallic plugs, and diameter "U' (U prime min)" is for composite plugs.
16. Red band location is critical. Dimensional verification of red band location is performed in Group A Inspection.
17. Threads are modified 60° stub configuration.
18. Gauges for plated threads: Maximum gauging limits of class 2A threads which are plated, are increased by the allowance. Unplated threads, for example class K, may be checked with standard after plate gauges.
19. Radius shall be 0.4 mm (.016 in) minimum, or optional chamfer shall be 0.18 mm (.007 in) X 45° ± 5° angle minimum. When the receptacle is mated to the plug, the radius or chamfer shall neither damage the plug's EMI band nor impede EMI sealing.
20. For shell size code used for part numbering, see table I.

FIGURE 9. Connector intermateability dimensions - Continued.

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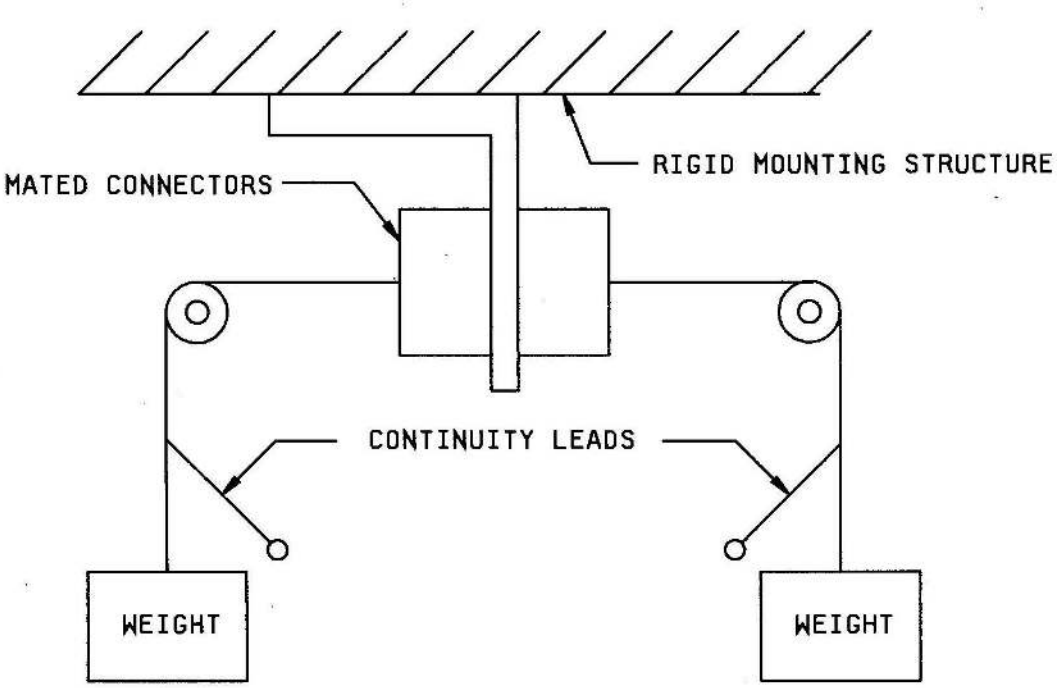


FIGURE 10. Typical fixturing for temperature life with contact loading (orientation optional).

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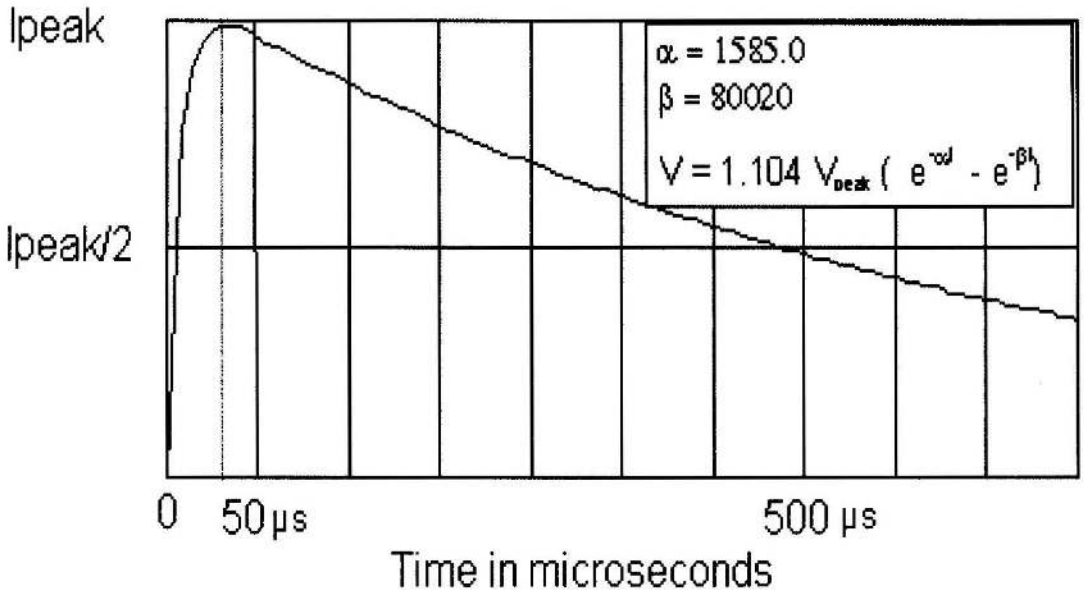


FIGURE 11. Waveform for indirect lightning strike effects.

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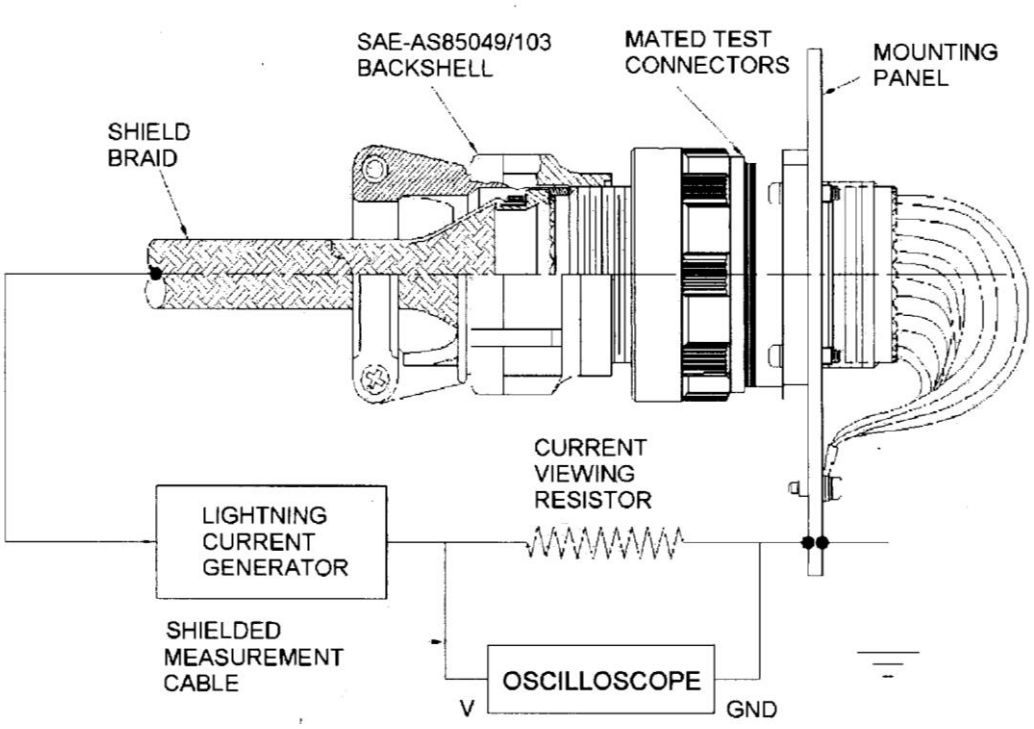


FIGURE 12. Indirect lightning strike test setup.

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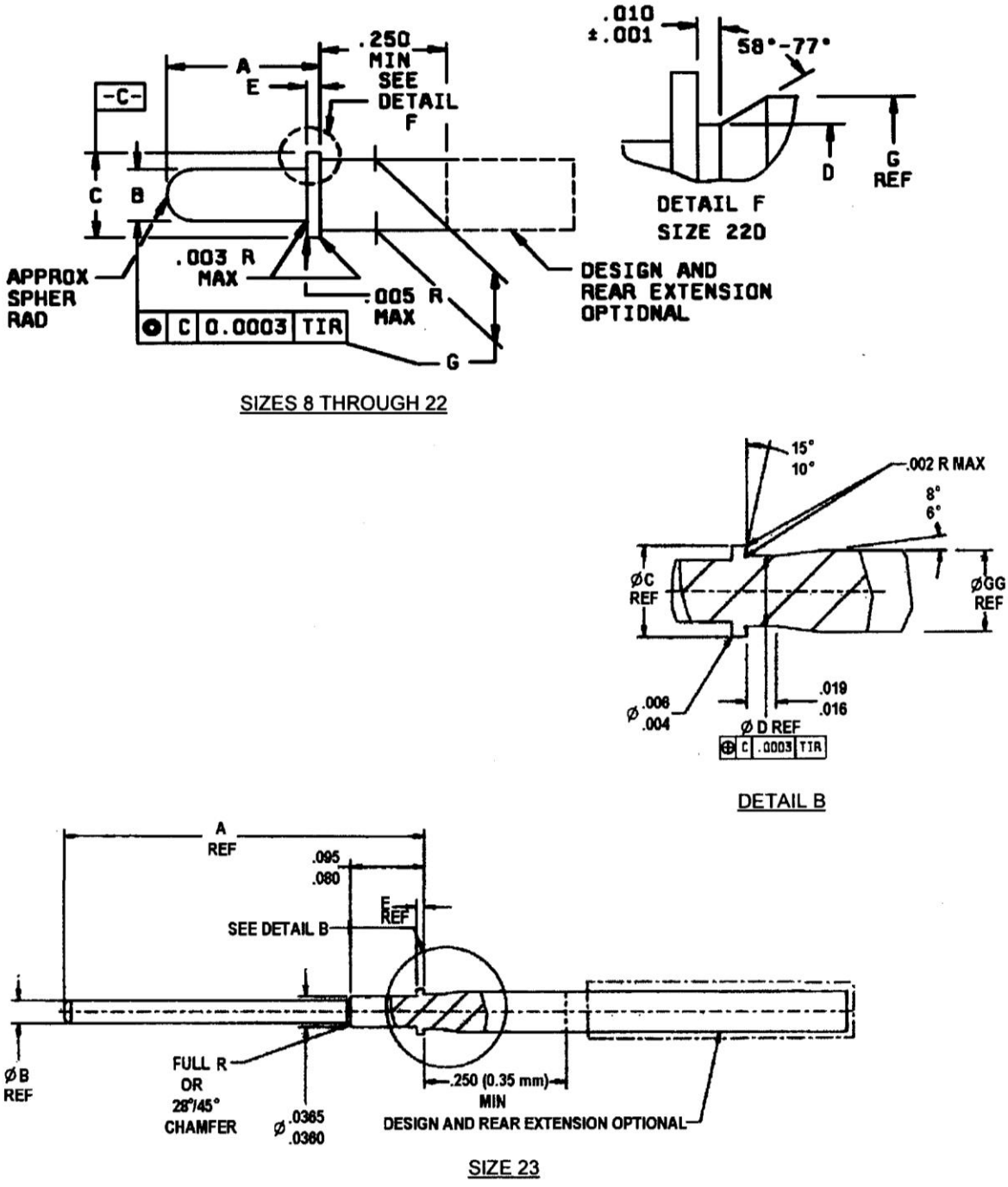


FIGURE 13. Test gauge, pin, and gauge location.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:58Z
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Contact size	A +.0005 -.0000	B dia +.0002 -.0000	C dia +.0002 -.0000	D dia +.0000 -.0002	E +.0000 -.0002	G dia +.0000 -.0002
23	.4260	.0265	.0540	.0415	.0090	.0480
22D	.3000	.0295	.0600	.0410	.0290	.0480
20	"	.0390	.0910	---	"	.0700
16	"	.0615	.1270	---	"	.1030
12	"	.0930	.1790	---	"	.1510
10	"	.1240	.2380	---	"	.2130
8 twinax or coax	"	.2170	.3110	---	"	.2760

Inches	mm	Inches	mm	Inches	mm
.0002	0.005	.0410	1.041	.1510	3.835
.0003	0.008	.0480	1.219	.1790	4.547
.0005	0.013	.0600	1.524	.2130	5.41
.001	0.03	.0615	1.562	.2170	5.51
.003	0.08	.0700	1.778	.2380	6.05
.005	0.13	.0910	2.311	.250	6.35
.010	0.25	.0930	2.362	.2760	7.01
.0290	0.737	.1030	2.616	.3000	7.620
.0295	0.749	.1240	3.15	.3110	7.899
.0390	0.991	.1270	3.226		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Material: Hardened tool steel.
4. Finish: 32 microinches polished.

FIGURE 13. Test gauge, pin, gauge location - Continued.

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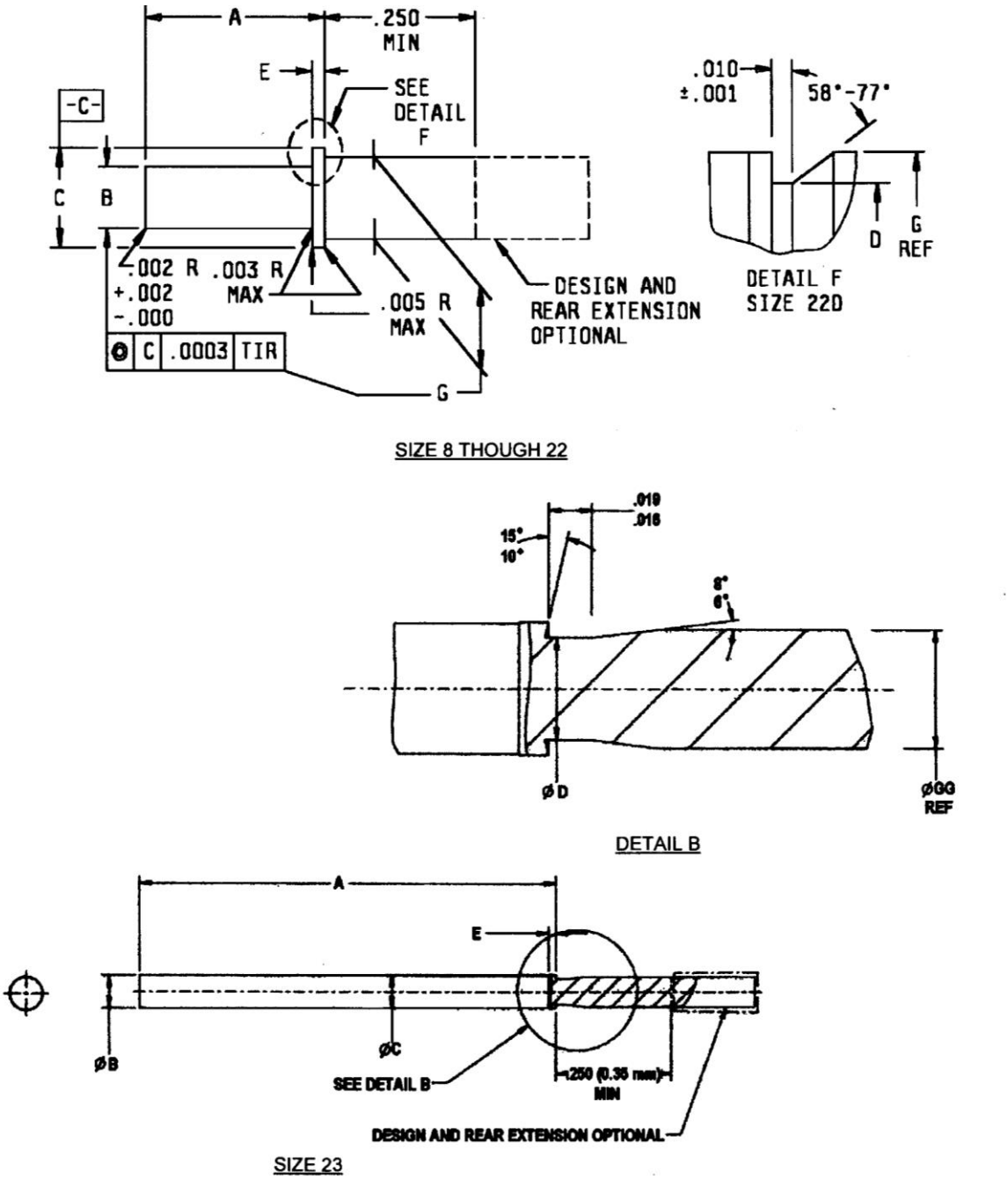


FIGURE 14. Test gauge, socket, gauge location.

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Contact size	A +.0005 -.0000	B dia +.0002 -.0000	C dia +.0002 -.0000	D dia +.0000 -.0002	E +.0000 -.0002	G dia +.0000 -.0002
23	.6780	.0540	.0540	.0415	.0120	.0480
22D	.5850	.0600	.0600	.041	---	.0480
20	"	.0770	.0910	---	.0330	.0700
16	"	.1120	.1270	---	"	.1030
12	"	.1600	.1790	---	"	.1510
10	"	.2190	.2380	---	"	.2130
8 twinax or coax	.5900	.2870	.3110	---	"	.2760

Inches	mm	Inches	mm	Inches	mm
.0002	0.005	.0480	1.219	.1790	4.547
.0003	0.008	.0600	1.524	.2130	5.41
.0005	0.013	.0700	1.778	.2190	5.56
.001	0.03	.0770	1.956	.2380	6.05
.002	0.05	.0910	2.311	.250	6.35
.003	0.08	.1030	2.616	.2760	7.01
.005	0.13	.1120	2.845	.2870	7.29
.010	0.25	.1270	3.226	.5850	14.859
.0330	0.838	.1510	3.835	.5900	14.99
.0410	1.041	.1600	4.064		

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Material: Hardened tool steel.
4. Finish: 32 microinches polished.
5. Dimension A conforms to point of spring engagement (C and Y, figures 1, 3, and 4).

FIGURE 14. Test gauge, socket, and gauge location - Continued.

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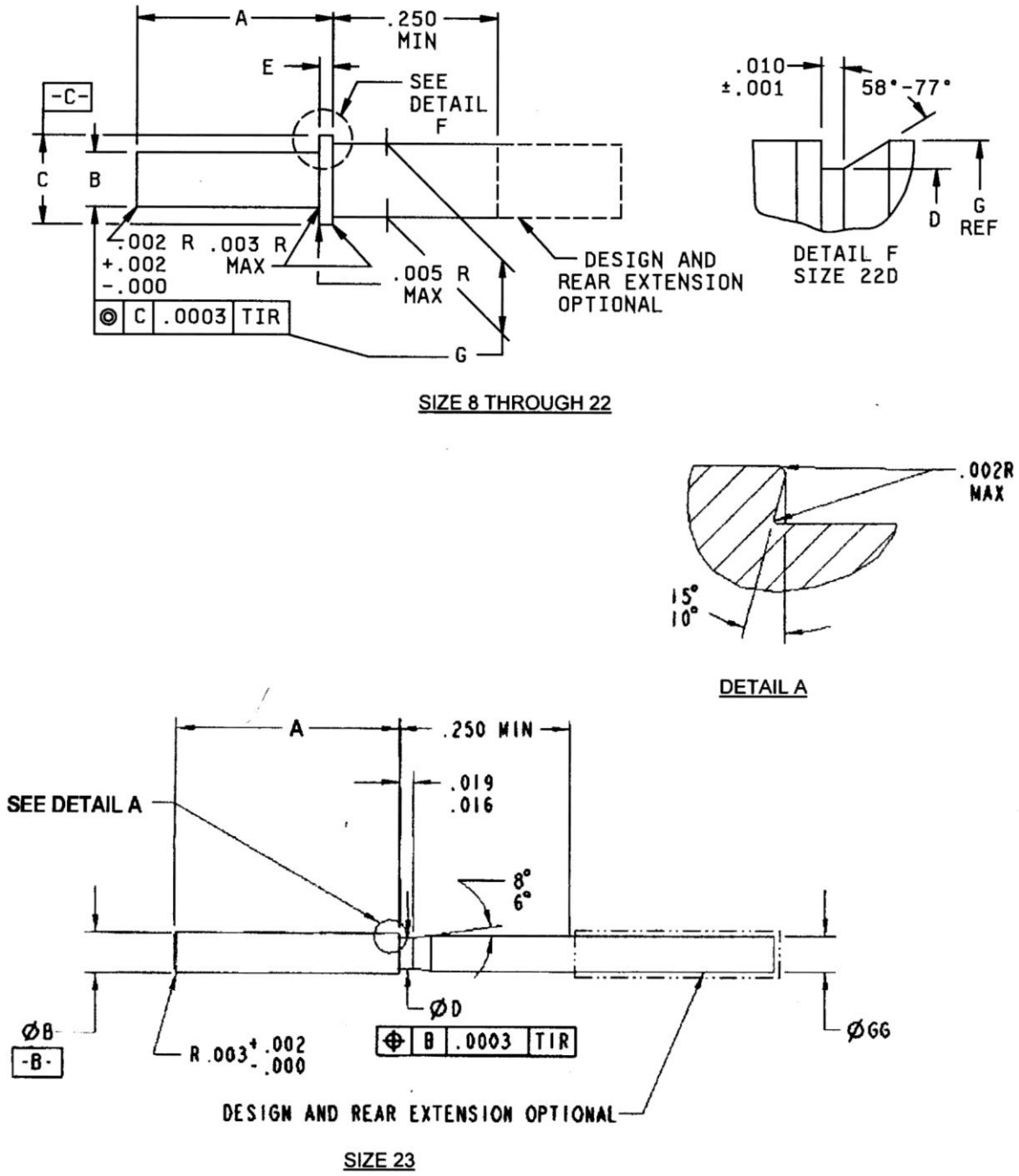


FIGURE 15. Test gauge, socket, and retention.

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Contact size	A +.0005 -.0000	B dia +.0002 -.0000	C dia +.0002 -.0000	D dia +.0000 -.0002	E +.0000 -.0002	G dia +.0000 -.0002
23	.7260	.0540	-----	.0415	----	.0480
22D	.6260	.0600	.0600	.0410	---	.0480
20	"	.0770	.0910	---	.0330	.0700
16	"	.1120	.1270	---	"	.1030
12	"	.1600	.1790	---	"	.1510
10	"	.2190	.2380	---	"	.2130
8 twinax or coax	.6310	.2870	.3110	---	"	.2760

Inches	mm	Inches	mm	Inches	mm
.0002	0.005	.0480	1.219	.1790	4.547
.0003	0.008	.0600	1.524	.2130	5.41
.0005	0.013	.0700	1.778	.2190	5.56
.001	0.03	.0770	1.956	.2380	6.05
.002	0.05	.0910	2.311	.250	6.35
.003	0.08	.1030	2.616	.2760	7.01
.005	0.13	.1120	2.845	.2870	7.29
.010	0.25	.1270	3.226	.3110	7.90
.0330	0.838	.1510	3.835	.6260	15.900
.0410	1.041	.1600	4.064	.6310	16.03

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Material: Hardened tool steel.
4. Finish: 32 microinches polished.

FIGURE 15. Test gauge, socket, and retention - Continued.

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CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – AS
Air Force – 85
DLA – CC

Preparing activity:
Navy - AS
Project 5935-2018-028

Review activities:
Army – AR, MI
Navy – EC, MC, OS
Air Force – 19, 99
MISC – MDA

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